3 to 16 Cells High Accuracy Battery Monitor and Protector

1 Features

- Highly integrated design to enhance system performance and reliability
  - Supports voltage monitoring for a range of 3 to 16 cells, addressing diverse application needs.
  - Cell voltage measurement range covers 1.5V to 5.0V, compatible with various battery types
  - Support for simultaneous current and voltage sampling
  - ±200 mV current sensing voltage range
  - Supports odd / even cell balancing for up to 8 cells with built-in balancing MOSFET (R_{ON} = 25Ω)
  - Integrated charge pump for high - side N-MOSFET control
  - Communication options include 400kHz I2C and 2MHz SPI
  - 2-channel programmable LDOs for external system usage
  - Integrated one-time-programmable (OTP) memory programmable by customers on production line
- Precise monitoring capability the full temperature range
  - Cell voltage measurement accuracy <7.5mV
  - Current sensing voltage measurement accuracy <50μV ($|V_{SRP}-V_{SRN}|$≤5mV)
  - Current sensing voltage measurement accuracy <1% ($|V_{SRP}-V_{SRN}|$>5mV)
  - Temperature measurement accuracy <2℃
- Operates safely and reliably in harsh environments
  - Supports voltage up to 85 V on cell connection pins, supports random cell attach sequence
  - Built-in chemical fuse driver for secondary protection
  - Extensive protection suite including voltage, temperature, current, and internal diagnostics
  - Provides multi-level fault alert and status indication functions
- Supports multiple power-saving operating modes
  - Sleep mode: 60μA
  - DeepSleep mode: 20μA
  - ShutDown mode: 5μA

2 Applications

- Battery Backups and Uninterruptible Power Supplies
- Energy Storage Systems
- E-Bikes and E-Scooters
- Power and Gardening Tools
- Other industrial battery packs

3 Description

The GD30BM2016 is a highly integrated battery monitor and protector designed for use in lithium-ion battery packs. It can accurately monitor the cell voltage, current, and temperature, while supporting high-side N-MOSFET control. The device integrates a high-precision monitoring system and a highly configurable protection subsystem, ensuring monitoring accuracy and simplifying system design. By providing overcharge and overdischarge protection, the GD30BM2016 effectively enhances battery safety and performance.

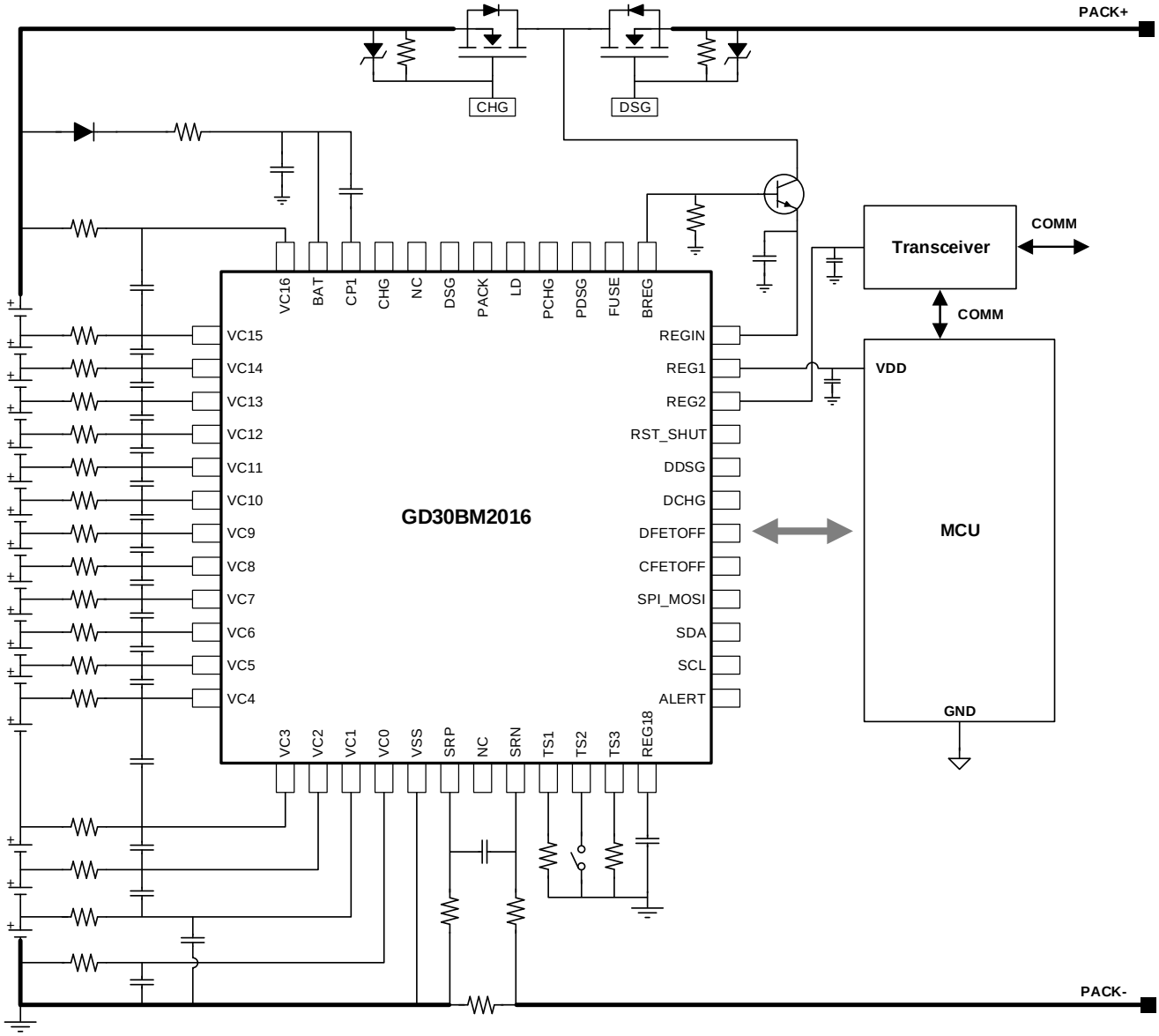
Device Information[1]

ORDERING CODE	PACKAGE	BODY SIZE (NOM)
GD30BM2016	LQFP-48	7.00 mm x 7.00 mm

1. For packaging details, see *Package Information* section.

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Simplified Circuit Schematic



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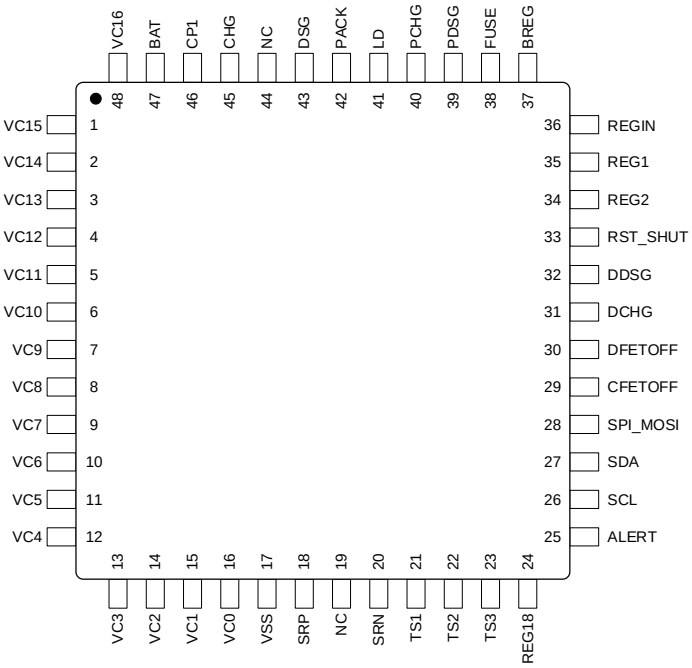
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4 Device Overview

4.1 Pinout and Pin Assignment

LQFP-48
Top View



4.2 Pin Description

PINS		TYPE[1]	FUNCTION
NAME	NUM		
VC15	1	I	Positive input pin of cell 15, balancing current input pin of cell 15, and balancing current return pin of cell 16
VC14	2	I	Positive input pin of cell 14, balancing current input pin of cell 14, and balancing current return pin of cell 15
VC13	3	I	Positive input pin of cell 13, balancing current input pin of cell 13, and balancing current return pin of cell 14
VC12	4	I	Positive input pin of cell 12, balancing current input pin of cell 12, and balancing current return pin of cell 13
VC11	5	I	Positive input pin of cell 11, balancing current input pin of cell 11, and balancing current return pin of cell 12
VC10	6	I	Positive input pin of cell 10, balancing current input pin of cell 10, and balancing current return pin of cell 11
VC9	7	I	Positive input pin of cell 9, balancing current input pin of cell 9, and balancing current return pin of cell 10

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PINS		TYPE[1]	FUNCTION
NAME	NUM		
VC8	8	I	Positive input pin of cell 8, balancing current input pin of cell 8, and balancing current return pin of cell 9
VC7	9	I	Positive input pin of cell 7, balancing current input pin of cell 7, and balancing current return pin of cell 8
VC6	10	I	Positive input pin of cell 6, balancing current input pin of cell 6, and balancing current return pin of cell 7
VC5	11	I	Positive input pin of cell 5, balancing current input pin of cell 5, and balancing current return pin of cell 6
VC4	12	I	Positive input pin of cell 4, balancing current input pin of cell 4, and balancing current return pin of cell 5
VC3	13	I	Positive input pin of cell 3, balancing current input pin of cell 3, and balancing current return pin of cell 4
VC2	14	I	Positive input pin of cell 2, balancing current input pin of cell 2, and balancing current return pin of cell 3
VC1	15	I	Positive input pin of cell 1, balancing current input pin of cell 1, and balancing current return pin of cell 2
VC0	16	I	Negative input pin of cell 1, balancing current return pin of cell 1
VSS	17	G	Ground
SRP	18	I	Positive sense pin, connected to the internal coulomb counter peripheral for integrating a small voltage between SRP and SRN, where SRP is the top of the sense resistor. A charging current generates a positive voltage at SRP relative to SRN
NC	19		Reserved for OTP Trim power pin, connect a 2.2 μF capacitor; when OTP is not used, no connection
SRN	20	I	Negative sense pin connected to the internal coulomb counter peripheral for integrating a small voltage between SRP and SRN, where SRN is the bottom of the sense resistor. A charging current generates a positive voltage at SRP relative to SRN.
TS1	21	I/O	Thermistor input, or general ADC input
TS2	22	I/O	Thermistor input and functions as wakeup from ShutDown, or general ADC input
TS3	23	I/O	Thermistor input, or general ADC input
REG18	24	O	Internal 1.8V LDO output (only for internal use)
ALERT	25	I/O	Multifunction pin, can be ALERT output, or thermistor input, or general ADC input, or general digital output
SCL	26	I/O	Multifunction pin, can be SCL or SPI_SCLK
SDA	27	I/O	Multifunction pin, can be SDA or SPI_MISO
SPI_MOSI	28	I/O	Multifunction pin, can be SPI_MOSI or I2C address, thermistor input, general ADC input, or general digital output
CFETOFF	29	I/O	Multifunction pin, can be CFETOFF, SPI_CS, thermistor input, general ADC input, or general digital output

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PINS		TYPE[1]	FUNCTION
NAME	NUM		
DFETOFF	30	I/O	Multifunction pin, can be DFETOFF, BOTHOFF, thermistor input, general ADC input, or general digital output
DCHG	31	I/O	Multifunction pin, can be DCHG, thermistor input, general ADC input, or general digital output
DDSG	32	I/O	Multifunction pin, can be DDSG, thermistor input, general ADC input, or general digital output
RST_SHUT	33	I	Digital input pin for reset or shutdown
REG2	34	O	Second LDO (REG2) output, which can be programmed for 1.8 V, 2.5 V, 3.0 V, 3.3 V, or 5.0 V
REG1	35	O	First LDO (REG1) output, which can be programmed for 1.8 V, 2.5 V, 3.0 V, 3.3 V, or 5.0 V
REGIN	36	I	Input pin for REG1 and REG2 LDOs
BREG	37	O	Base control signal for external preregulator transistor
FUSE	38	I/O	Fuse sense and drive pin
PDSG	39	O	Predischarge P-MOSFET control pin
PCHG	40	O	Precharge P-MOSFET control pin
LD	41	I/O	Load detect pin
PACK	42	I	Pack sense input pin
DSG	43	O	Discharge N-MOSFET drive output pin
NC	44		Not connected
CHG	45	O	Charge N-MOSFET drive output pin
CP1	46	I/O	Charge pump capacitor
BAT	47	I	Power supply input pin
VC16	48	I	Positive input pin of cell 16, balancing current input pin of cell 16, and top-of-stack measurement point

1. I = Input, O = Output, P = Power, G = Ground.

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5 Parameter Information

5.1 Absolute Maximum Ratings[1]

over operating free-air temperature range (unless otherwise noted)

SYMBOL	PARAMETER	MIN	MAX	UNIT
V_{BAT}, V_{PACK}, V_{LD}, V_{CHG}, V_{DSG}	BAT, PACK, LD, CHG, DSG to VSS	-0.3	85	V
V_{PCHG}, V_{PDSG}	PCHG, PDSG to VSS	Max (V_{BAT}, V_{LD})	85	V
VCn	VCn to VSS, n=1~16	VC(n-1)-0.3	85	V
VC0	VC0 to VSS	-0.3	5	V
V_{SCL}, V_{SDA}, $V_{SPI-MOSI}$, $V_{CFETOFF}$, $V_{DFETOFF}$, V_{RST_SHUT} (when used as digital input)	SCL, SDA, SPI-MOSI, CFETOFF, DFETOFF, RST_SHUT to VSS	-0.3	6	V
V_{TS1}, V_{TS2}, V_{TS3}, V_{ALERT}, $V_{CFETOFF}$, $V_{DFETOFF}$, $V_{SPI-MOSI}$, V_{DCHG}, V_{DDSG} (when used as thermistor or general ADC input)	TS1, TS2, TS3, ALERT, CFETOFF, DFETOFF, SPI-MOSI, DCHG, DDSG to VSS	-0.3	VREG18+0.3	V
V_{REGIN}	REGIN to VSS	Max (-0.3, VBREG-5.5)	MIN (6, VBREG+0.3)	V
V_{BREG}	BREG to VSS	Max (-0.3, VREGIN-0.3)	VREGIN+5.5	V
V_{REG18}	REG18 to VSS	-0.3	2	V
V_{REG1}, V_{REG2}	REG1, REG2 to VSS	-0.3	MIN (6, VREGIN+0.3)	V
V_{SRP}, V_{SRN}	SRP, SRN to VSS	-0.3	VREG18+0.3	V
V_{TS2} (for wakeup), V_{ALERT}, $V_{CFETOFF}$, $V_{DFETOFF}$, V_{SPI_MOSI}, V_{DCHG}, V_{DDSG} (when used as digital output)	TS2, ALERT, CFETOFF, DFETOFF, SPI_MOSI, DCHG, DDSG to VSS	-0.3	6	V
V_{CP1}	CP1 to VSS	VBAT-0.3	MIN (85, VBAT+15)	V
V_{FUSE}	FUSE to VSS	-0.3	MIN (20, VBAT+0.3)	V
I_{CB}	Maximum cell balancing current through a single cell		80	mA
T_A	Operating ambient temperature	-40	+125	℃
T_J	Junction temperature	-40	+150	℃
T_{STG}	Storage temperature	-40	+150	℃
T_L	Reflow soldering, 10s		+260	℃

1. Operation outside the Absolute Maximum Ratings may cause permanent device damage. Absolute Maximum Ratings do not imply functional operation of the device at these or any other conditions beyond those listed under Recommended Operating Conditions.

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5.2 Recommended Operating Conditions

SYMBOL	PARAMETER	MIN	TYPE	MAX	UNIT
V_{BAT}	Supply operating voltage range	8		80	V
V_{PACK} and V_{LD}	PACK, LD operating voltage range	0		80	V

5.3 ESD Ratings

SYMBOL	PARAMETER	VALUE	UNIT
$V_{ESD(HBM)}$	Human body model (HBM), ANSI/ESDA/JEDEC JS-001-2017[1]	±2000	V
$V_{ESD(CDM)}$	Charged device model (CDM), ANSI/ESDA/JEDEC JS-002-2022[2]	±500	V

1. Except LD Pin，LD Pin ±1000V pass；JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
2. JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.4 Thermal Resistance[1]

SYMBOL	CONDITIONS	LQFP48	UNIT
θ_{JA}	Junction-to-ambient thermal resistance	65	℃/W
θ_{JC}	Junction-to-case thermal resistance	19	℃/W

1. Metrics provided using set-up conditions compliant with EIA/JESD51-2, 7, and 8.

5.5 Electrical Characteristics

16 cells, each cell voltage = 3.7V, V_{BAT} = 59.2V, TA = 25℃, unless otherwise noted.

SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
Current Measurement						
t_{IADC}	Current ADC conversion time			1/3		ms
V_{SRPN}	Current ADC measurement range	T_A = -40℃ to +85℃	-200		+200	mV
$I_{ADC_GAIN_ERR}$	Current ADC measurement error	$\|V_{SRP} - V_{SRN}\|$ > 5mV, T_A = -40℃ to +85℃	-1		+1	%
V_{IADC_OFFSET}	Current sensing offset	$\|V_{SRP} - V_{SRN}\|$ ≤ 5mV, T_A = -40℃ to +85℃	-50		+50	μV
I_{SRPN_LEAK}	SRP and SRN leakage	IADC not running, T_A = -40℃ to +85℃	-500		+500	nA
I_{SRPN_DIFF}	SRP and SRN differential input current	IADC running, $\|V_{SRP} - V_{SRN}\|$ = 200mV, TA = -40℃ to +85℃		1		μA
		IADC running, $\|V_{SRP} - V_{SRN}\|$ < 5mV, TA = -40℃ to +85℃		50		nA
Cell Voltage Measurement						
t_{VADC}	Cell voltage conversion time	16 bits, T_A = 25℃		1		ms

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
V_{CELL}	Cell ADC measurement range	T_A = -40℃ to +85℃	1.5		5	V
V_{CELL_ERR}	Total cell measurement error	V_{CELL}=2V to 4.5V, T_A = 25℃	-3		+3	mV
		V_{CELL}=2V to 4.5V, T_A =-40℃ to +85℃	-7.5		+7.5	mV
Temperature Measurement						
T_{INT_ERR}	Internal temperature measurement error	T_A = -40℃ to 85℃	-2		+2	℃
V_{NTC_ERR}	NTC total measurement error	T_A = -40℃ to 85℃	-2		+2	℃
$I_{NTC_IN_LEAK}$	NTCx input leakage	ADC off, on each NTC pin, T_A = -40℃ to +85℃			250	nA
$I_{NTC_LEAK_CONV}$	NTCx input leakage ADC on	ADC converting NTCx, on each NTC pin			250	nA
STACK Measurement						
V_{STACK}	Stack voltage measurement range	T_A = -40℃ to +85℃	8		80	V
V_{STACK_ERR}	Stack voltage measurement error	V_{VC16} = 0~80V， T_A = -40℃ to +85℃	-0.3		+0.3	V
LD Measurement						
V_{LD}	LD voltage measurement range	T_A = -40℃ to +85℃	0		80	V
V_{LD_ERR}	LD voltage measurement error	V_{LD}=0~80V, T_A = -40℃ to +85℃	-0.3		+0.3	V
PACK Measurement						
V_{PACK}	Pack voltage measurement range	T_A = -40℃ to +85℃	0		80	V
V_{PACK_ERR}	Pack voltage measurement error	V_{PACK}=0~80V, T_A = -40℃ to +85℃	-0.3		+0.3	V
Hardware Protection (COV/CUV)						
$V_{CELL_TH_STEP}$	Cell OV/UV steps	V_{CELL} = 1.5V to 5V, T_A = 25℃		20		mV
$V_{CELL_TH_RANGE}$	Cell OV/UV setting range		1.5		5.0	V
$V_{CELL_TH_ERR}$	Cell OV/UV Detection error	V_{CELL} = 1.5V to 5V, T_A = 25℃	-10		+10	mV
		V_{CELL} = 1.5V to 5V, T_A = -40℃ to 85℃	-25		+25	mV
V_{HYS}	Cell OV/UV Hysteresis Range		0		1	V
$t_{CELL_DLY_STEP}$	Cell OV/UV detection delay step	N is the number of selected cells		0.186*(N+5)		ms
$n_{CELL_DLY_RANGE}$	Cell OV/UV detection delay range	V_{CELL} = 1.5V to 5V, T_A = 25℃	1		127	Step

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
Hardware Protection (SCD)						
V_{SCD}	SCD voltage threshold Nominal settings	Threshold based on V_{SRP} - V_{SRN}	-10/-20/-40/-60/-80/-100/-125/-150/-175/-200/-250/-300/-350/-400/-450/-500			mV
V_{SCD_ERR}	SCD voltage threshold detection error	$\|V_{SCD}\|$ ≤ 20mV, T_A = -40℃ to 85℃	-5		+5	mV
		$\|V_{SCD}\|$ > 20mV, T_A = -40℃ to 85℃	-20		+20	%
$t_{SCD_DLY_STEP}$	SCD detection delay steps			32		μs
$t_{SCD_DLY_RANGE}$	SCD detection delay range		0		992	μs
t_{SCD_PD}	SCD Comparator propagation delay	V_{SRN} - V_{SRP} Input overdrive = 25mV		8		μs
Hardware Protection（OCC/OCD1/OCD2）						
V_{OCD}	OCD1/OCD2 voltage threshold range		-200		-4	mV
V_{OCC}	OCC voltage threshold range		4		124	mV
V_{OC_STEP}	OCC/OCD1/OCD2 voltage threshold steps			2		mV
V_{OC_ERR}	OCC/OCD1/OCD2 voltage threshold detection error	$\|V_{OC}\|$<20 mV, T_A = -40℃ to 85℃	-2.5		+2.5	mV
		$\|V_{OC}\|$=20 ~ 60 mV, T_A = -40℃ to 85℃	-4		+4	mV
		$\|V_{OC}\|$=60 ~ 100 mV, T_A = -40℃ to 85℃	-5		+5	mV
		$\|V_{OC}\|$ >100 mV, T_A = -40℃ to 85℃	-6		+6	mV
$t_{OC_DLY_STEP}$	OCC/OCD1/OCD2 detection delay steps	N is the number of selected cells		0.186*(N+5)		ms
$n_{OC_DLY_RANGE}$	OCC/OCD1/OCD2 detection delay range		1		127	Step
Power Supply						
V_{BAT}	Supply voltage range		8		80	V
V_{POR}	Power-on reset	BAT voltage rising	4	5	6	V
V_{POR_HYS}	Power-on reset hysteresis	BAT voltage falling		0.2		V
I_{NORMAL}	Normal mode current	Regular measurements and protections active, REG1 = 3.3 V with no load, REG2 OFF, CHG/DSG ON in 11V overdrive mode, no communication		960		μA

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
I_{SLEEP}	Sleep mode current	Periodic protections and monitoring, no pack current, REG1 / REG2 OFF, CHG OFF, DSG source follower mode, no communication, Voltage_Time = 5s		60		μA
$I_{DEEPSLEEP}$	DeepSleep mode current	No monitoring or protections, REG0 / REG1 / REG2 = OFF, no communication		20		μA
$I_{SHUTDOWN}$	ShutDown mode current	All blocks powered down, with the exception of the TS2 wakeup circuit, no monitoring or protections, no communication		5		μA
Voltage References						
V_{REF1}	Reference voltage 1	T_A=25℃	1.2115	1.212	1.2125	V
$V_{REF1DRIFT}$	Reference voltage 1 drift	T_A=-40℃ to 85℃		±6		ppm/℃
V_{REF2}	Reference voltage 2	T_A=25℃	1.1923	1.2	1.2077	V
$V_{REF2DRIFT}$	Reference voltage 2 drift	T_A=-40℃ to 85℃		±24		ppm/℃
LDO REG18						
V_{REG18}	REG18 voltage	T_A=-40℃ to 85℃	1.7	1.8	1.9	V
$I_{REG18SC}$	REG18 short-circuit current limit	T_A=-40℃ to 85℃	5.2	6.3	7.4	mA
Pre-LDO REG0						
V_{REGIN_INT}	Pre-LDO voltage, when generated using BREG		5.0	5.5	5.8	V
V_{REGIN_EXT}	Pre-LDO voltage, when using externally supplied REGIN				5.5	V
I_{BREGSC}	BREG short-circuit current limit			5.4		mA
C_{EXT}	External capacitor REGIN to VSS		11	22	33	nF
LDO REG1/REG2						
$V_{REG1/2(1.8)}$	REG1/2 output voltage, nominal 1.8V setting	I_{LOAD} = 0 mA to 45 mA, T_A=-40℃ to 85℃	1.7	1.8	1.9	V
$V_{REG1/2(2.5)}$	REG1/2 output voltage, nominal 2.5V setting	I_{LOAD} = 0 mA to 45 mA, T_A=-40℃ to 85℃	2.3	2.5	2.7	V
$V_{REG1/2(3.0)}$	REG1/2 output voltage, nominal 3.0V setting	I_{LOAD} = 0 mA to 45 mA, T_A=-40℃ to 85℃	2.8	3.0	3.2	V
$V_{REG1/2(3.3)}$	REG1/2 output voltage, nominal 3.3V setting	I_{LOAD} = 0 mA to 45 mA, T_A=-40℃ to 85℃	3.1	3.3	3.5	V

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
$V_{REG1/2(5.0)}$	REG1/2 output voltage, nominal 5.0V setting	I_{LOAD} = 0 mA to 45 mA, T_A=-40℃ to 85℃	4.5	5.0	5.3	V
$I_{REG1/2SC}$	REG1/2 short-circuit current limit	T_A=-40℃ to 85℃	52	62	80	mA
C_{REG1}	External capacitor REG1/2 to VSS		1	2.2	4.7	μF
LD Pin						
I_{LD_PU}	Internal pullup current from BAT pin to LD pin	Used for load detect functionality, V_{LD}=VSS	35	100	172	μA
R_{LD_PD}	Internal pulldown resistance on LD pin	ShutDown mode		90		kΩ
FUSE Pin						
V_{OH}	Output voltage high (when driving fuse)		6	7	9	V
V_{IH}	High-level input (for fuse detection)		2		12	V
V_{IL}	Low-level input (for fuse detection)				0.7	V
Thermistor Measurement						
R_{TS_PU}	Internal pullup resistance	Setting for nominal 18kΩ		18.8		kΩ
R_{TS_PAD}	Internal pad resistance			427		Ω
CHG/DSG Driver						
V_{CHG_ON}	CHG ON, CHG to BAT voltage	C_L = 20nF, charge pump low overdrive setting	4.5	5.0	7.0	V
		C_L = 20nF, charge pump high overdrive setting	9	11	13	V
V_{DSG_ON}	DSG ON, DSG to BAT voltage	C_L = 20nF, charge pump low overdrive setting	4.5	5.0	7.0	V
		C_L = 20nF, charge pump high overdrive setting	9	11	13	V
V_{DSG_Source}	DSG source follower, DSG to BAT voltage	C_L = 20nF, source follower mode		0		V
V_{CHG_OFF}	CHG OFF, CHG to BAT voltage	C_L = 20nF			0.7	V
V_{DSG_OFF}	DSG OFF, DSG to LD voltage	C_L = 20nF			0.7	V
t_{CHG_Rise}	CHG rise time	C_L = 20nF, R_{GATE} = 100Ω, 10% to 90% of V_{CHG_ON}, charge pump high overdrive setting		18		μs

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
t_{CHG_Fall}	CHG fall time to BAT	C_L = 20nF，R_{GATE} = 100Ω，90% to 10% of V_{CHG_ON}, charge pump high overdrive setting		20		μs
t_{DSG_Rise}	DSG rise time	C_L = 20nF，R_{GATE} = 100Ω, 10% to 90% of V_{DSG_ON}, charge pump high overdrive setting		18		μs
t_{DSG_Fall}	CHG fall time to LD	C_L = 20nF，R_{GATE} = 100Ω，90% to 10% of V_{DSG_ON}, charge pump high overdrive setting		2		μs
R_{CHG_ON}	CHG gate drive turn-on resistance			400		Ω
R_{CHG_OFF}	CHG gate drive turn-off resistance			400		Ω
R_{DSG_ON}	DSG gate drive turn-on resistance			900		Ω
R_{DSG_OFF}	DSG gate drive turn-off resistance			900		Ω
t_{CP_START}	Charge pump start up time	C_L = 20nF，C_{CP1}= 470nF，10% to 90% of V_{CHG_ON}		100		ms
C_{CP1}	Charge pump capacitor		100	470	2200	nF
PCHG/PDSG Driver						
V_{PCHG_ON}	PCHG ON	Max (V_{PACK}, V_{BAT}) - V_{PCHG}	7.5	8.5	9.5	V
V_{PDSG_ON}	PDSG ON	Max (V_{LD}, V_{BAT}) - V_{PDSG}	7.5	8.5	9.5	V
I_{PD}	Current sink capability，PCHG and PDSG			25		μA
Cell Open Wire Detector						
I_{OW}	Internal cell open wire check current		66	168	285	μA
Cell Balancing						
R_{CB}	Internal cell balancing resistance	$R_{DS(ON)}$ for internal FET switch at $V_{VC(n)}$ -$V_{VC(n-1)}$ = 3V, 1 ≤ n ≤ 16	15	25	40	Ω
Internal Oscillators						
f_{HFO}	High-frequency oscillator	V_{BAT} = 8V to 80V		2		MHz
f_{HFO_ERR}	High-frequency oscillator error	V_{BAT} = 8V to 80V, T_A= -40℃ to +85℃	-4		4	%
f_{LFO}	Low-frequency oscillator	V_{BAT} = 8V to 80V		768		kHz
f_{LFO_ERR}	Low-frequency oscillator error	V_{BAT} = 8V to 80V, T_A = -40℃ to +85℃	-2.5		2.5	%
Digital I/O						
V_{IH}	High-level input	SCL, SDA, SPI_MOSI, CFETOFF, DFETOFF, RST_SHUT	0.66* V_{REG18}			V

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SYMBOL	PARAMETER	TEST CONDITIONS	MIN	TYPE	MAX	UNIT
V_{IL}	Low-level input	SCL, SDA, SPI_MOSI, CFETOFF, DFETOFF, RST_SHUT			0.33* V_{REG18}	V
V_{OH}	Output voltage high, TS2	TS2 during ShutDown mode	4.5			V
	Output voltage high, 5 V case	ALERT, SDA (configured as SPI_MISO), SCL (configured as SPI_SCLK), CFETOFF (configured as GPO), DFETOFF (configured as GPO), DCHG, DDSG, pins driving from REG1, V_{REG1} set to 5 V nominal setting	0.9*V_{REG1}			V
V_{OL}	Output voltage low, 5 V case	ALERT, SPI_MOSI, SCL, SDA, DCHG, DDSG, CFETOFF (configured as GPO), DFETOFF (configured as GPO), pins driving from REG1, V_{REG1} set to 5 V nominal setting			0.77	V
R_{RST_PD}	Internal pulldown resistance on RST_SHUT pin			200		kΩ
I2C Communication Interface						
f_{I2C}	I2C clock frequency	T_A = 25℃			400	kHz
V_{ILOW}	SCL/SDA low input voltage	T_A = -40℃ to +85℃			0.33 x V_{REG18}	V
V_{IHIGH}	SCL/SDA high input voltage	T_A = -40℃ to +85℃	0.66* V_{REG18}			V
SPI Communication Interface						
f_{SPI}	SPI clock frequency	T_A = 25℃			2	MHz
$V_{SPI_IN_LOW}$	SPI input low	SDI, CLK, T_A = -40℃ to +85℃			0.33 x V_{REG18}	V
$V_{SPI_IN_HIGH}$	SPI input high	SDI, CLK, T_A = -40℃ to +85℃	0.66* V_{REG18}			V
$R_{SDO_DRV_HI}$	SDO high-side (HS) on resistance	I_{SOURCE} = 1.5mA, T_A = -40℃ to +85℃	75	100	125	Ω
$R_{SDO_DRV_LOW}$	SDO low-side (LS) on resistance	I_{SINK} = 1.5mA, T_A = -40℃ to +85℃	38	52	66	Ω

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6 Functional Description

6.1 Functional Block Diagram

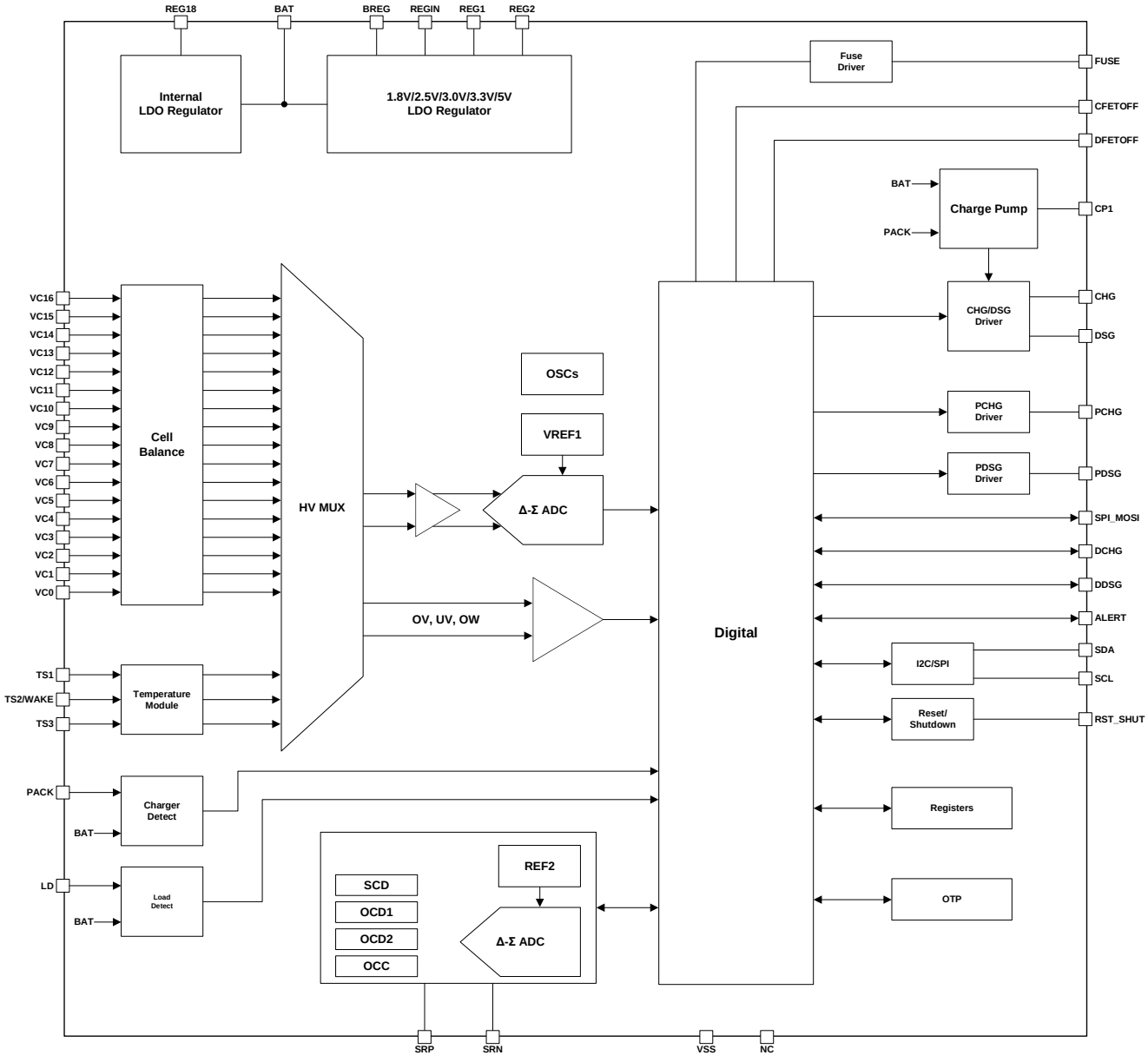


Figure 1. Functional Block Diagram

6.2 Overview

The GD30BM2016 is a highly integrated, accurate battery monitor and protector for 3-series to 16-series Li-ion, Li-polymer, and LiFePO4 battery packs. A high-accuracy voltage, current, and temperature measurement provides data for host-based algorithms and control. A feature-rich and highly configurable protection subsystem provides a wide set of protections that can be triggered and recovered completely autonomously by the device or under full control of a host processor.

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The GD30BM2016 integrates an independent 16‑bit ADC for current sensing, ensuring continuous current acquisition. It also incorporates current averaging and coulomb counting functions. The coulomb counter can operate in a low‑power Sleep mode, allowing it to continuously integrate small system currents even when the host MCU is in standby mode.

The integrated charge pump with high-side protection N-MOSFET drivers enables host communication with the device even when FETs are off by preserving the ground connection to the pack. Dual programmable LDOs are included for external system use, with each independently programmable to voltages of 1.8 V, 2.5 V, 3.0 V, 3.3 V, and 5.0 V, capable of providing up to 45 mA each.

The GD30BM2016 includes one-time-programmable (OTP) memory for customers to set up device operation on their own production line. Multiple communications interfaces are supported, including 400-kHz I2C, SPI. Multiple digital control and status data are available through several multifunction pins on the device, including an interrupt to the host processor, and independent controls for host override of each high-side protection N-MOSFET. Three dedicated pins enable temperature measurements using external thermistors, and multifunction pins can be programmed to use for additional thermistors, supporting a total of up to nine thermistors, in addition to an internal die temperature measurement.

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7 Measurement Subsystem

7.1 Measurement Subsystem Overview

The GD30BM2016 is capable of monitoring the following key parameters: cell voltage, current, thermistor temperature, VC16 pin voltage (V_{STACK}), PACK voltage (V_{PACK}), LD voltage (V_{LD}), and internal chip temperature. The measurement results of these voltage, current, and temperature parameters can be directly accessed by the user through command operations.

7.2 Voltage Measurement

The GD30BM2016 integrates a voltage ADC that is multiplexed between measurements of cell voltages, an internal temperature sensor, and up to nine external thermistors, as well as performs measurements of the voltage at the VC16 pin, the PACK pin, the LD pin, the internal REG18 LDO voltage.

The GD30BM2016 device supports measurement of individual differential cell voltages in a series configuration, ranging from 3-series cells to 16-series cells. Each cell voltage measurement is a differential measurement of the voltage between two adjacent cell input pins, such as VC1-VC0, VC2-VC1, and so forth. The cell voltage measurements are processed based on trim and calibration corrections, and then reported in 16-bit resolution using units of userV. The raw 24-bit digital output of the ADC is also available for readout using 32-bit subcommands.

The GD30BM2016 supports measurement of the VC16, PACK, and LD voltages, as well as the internal chip temperature. The device integrates an internal resistor divider network that divides down the VC16, PACK, and LD voltages before feeding them into the internal ADC. The corresponding measurement results are reported in units of userV. This design reduces additional current consumption that would otherwise result from external resistor dividers and also prevents occupation of the processor's ADC channels.

If the GD30BM2016 is used in a system with fewer than 16-series cells, it is necessary to ensure that VC16, VC1, and VC0 are properly connected. In the **VCELL_MODE Register**, the bits corresponding to the actually connected cells must be set to 1, **VCELL_MODE[15:10]** corresponds to Cell16 ~ Cell11, **VCELL_MODE[8:0]** corresponds to Cell10 ~ Cell2, and **VCELL_MODE[9]** is reserved and should be fixed to 0. For any unused VC pins, they must be shorted to adjacent pins.

7.3 Current Measurement

The GD30BM2016 monitors pack current using a low-side sense resistor, which connects to the SRP and SRN pins through an external RC filter, which should be connected such that a charging current will create a positive voltage on SRP relative to SRN. The differential voltage between SRP and SRN is digitized by an integrated coulomb counter ADC, which can digitize voltages over a ±200 mV range and uses multiple digital filters to provide optimized measurement of the instantaneous, averaged, and integrated current. The device supports a wide range of sense resistor value, with a larger value providing better resolution for the digitized result. The maximum value of sense resistor should be limited to ensure the differential voltage remains within the ±200 mV range for system operation when current measurement is desired. For example, a system with maximum discharge current of 200 A during normal operation (not a fault condition) should limit the sense resistor to 1 mΩ or below.

The GD30BM2016 integrates a dedicated ADC channel for current measurement, ensuring continuous current

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detection, which guarantees the accuracy of both current measurement and coulomb counting calculation.

The GD30BM2016 supports three current measurement filtering modes and a coulomb counting function:

CC1 Filter Current Output:

- Generates a 16-bit current measurement, which is the average of fixed-length CC2 results, representing the average continuous current.
- Can be used by the main controller for SoC calculation.
- Produces one output every 100 ms in Normal mode.
- Must be accessed using the data indirect subcommand CC1 Current.

CC2 Filter Current Output:

- Generates a 16-bit current measurement for instantaneous current measurement.
- In Normal mode, with the default ADC divider, one output is generated every 1/3 ms, and the current value is reported in a 16-bit format.
- Provides 24-bit raw CC2 data for host use, transmitted in a 32-bit format (the lower 24 bits contain the data, and the upper 8 bits are sign-extended).
- The CC2 value can be accessed using the direct command CC2 Current.

CC3 Filter Current Output:

- The output is the average of a programmable number of CC2 current samples.
- The CC3 output is reported in a 16-bit format.
- It can be accessed using the data indirect subcommand CC3 Current.

Coulomb Counter Output:

- Represented as a 64-bit value, where the upper 32 bits indicate the integer part of the accumulated charge and the lower 32 bits indicate the fractional part.
- Includes an additional 32-bit time accumulation value, expressed in userAh units.
- Both the accumulated charge integration and timer can be reset by the host via commands sent through the digital communication interface.
- The Coulomb counter value can be accessed using the data indirect subcommands Accum Charge, Accum Charge Fraction, and Accum Time.
- The Coulomb counter can be reset using the non-data subcommand Reset PassQ to prevent overflow caused by long-term accumulation.

7.4 Synchronized Voltage and Current Measurement

While the cell voltages are digitized sequentially using a single muxed ADC during normal operation, the current is digitized continuously by the dedicated coulomb counter ADC. The current is measured synchronously with each cell voltage measurement, and can be used for individual cell impedance analysis. The ongoing periodic current measurements can be read out through the digital communication interface, while the measurements taken that were synchronized with particular cell voltage measurements are stored paired with the associated cell voltage measurement.

The synchronized voltage and current data can be accessed using the data indirect subcommands **Cell x Voltage Counts** and **Cell x Current Counts**, which ensures the synchronously aligned voltage and current data are read out together.

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7.5 Temperature Measurement

The GD30BM2016 integrates an internal temperature sensor that can be used to measure the device's own temperature. When the device operates in balancing mode, the internal MOSFETs generate heat, and the balancing algorithm determines whether to enable or disable balancing based on the internal chip temperature.

The GD30BM2016 includes an on-chip temperature measurement and can also support up to nine external thermistors on multifunction pins (TS1, TS2, TS3, CFETOFF, DFETOFF, ALERT, SPI_MOSI, DCHG, and DDSG). The device includes an internal pullup resistor to bias a thermistor during measurement. The internal pullup resistor has two options which can set the pullup resistor to either 18kΩ or 180kΩ (or none at all). The 18kΩ option is intended for use with thermistors such as the Semitec 103-AT, which has 10kΩ resistance at room temperature.

Users can configure through registers which pin is used for thermistor measurement. The pullup resistor value is configured uniformly in **PAD CFG: ts_res_sel**. The thermistor measurement result can be configured for cell or charge/discharge MOSFET temperature protection.

As shown in Table 1, the configuration of multifunction pins for temperature measurement is illustrated. If the user intends to use the CFETOFF pin for temperature measurement and for cell temperature protection, the high bits of the **CFETOFF Pin Config Register** should be configured as follows: **set cfetoff_opt OPT[4]** to 1 to disable digital output; set **OPT[3]** to 0 so it is not used as an ADC input; set **OPT[2]** to 1 to select the pullup resistor for the Tsensor input; set **OPT[1]** to 0; and set **OPT[0]** to 1 to assign it for cell protection. The **cfetoff_pin_fxn** should be set to 3 to select ADC/NTC input. All other unspecified bits can be set to 0. The **PAD CFG Register** is used to configure and enable the pullup resistor value.

Table 1. CFETOFF Pin Config Register

Field	Bits	Default	Type	Field Description
cfetoff_opt	[7: 2]	0x4	RW	The options for pin functions selected by these bit configurations are as follows: OPT[5]: Polarity when configured as CFETOFF (no effect on GPO mode) 0 = Select active high 1 = Select active low; OPT[4]: 0 = Digital output (OEN=0) 1 = No digital output (OEN=1, HZ) OPT[3]: 0 = Output high-level drive using REG1 1 = Output high-level drive using REG18 OPT[2]: 0 = Disable digital input (IE=0) 1 = Enable digital input (IE=1) OPT[1]: 0 = Disable digital input weak pull-up 1 = Enable digital input weak pull-up OPT[0]: 0 = Disable digital input weak pull-down

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Field	Bits	Default	Type	Field Description
				1 = Enable digital input weak pull-down When selecting pins for ADCIN or thermistor functions, the OPT[5:0] bits will be used as follows: OPT[5:4]: If unused, set OPT[4] to 1. OPT[3]: Whether the pin is used as ADC input: 0 = No 1 = Yes OPT[2]: Whether the pin is used as Tsensor input with pull-up resistor 0 = No 1 = Yes OPT[1]: 0 = Not used for FET temperature protection 1 = Used for FET temperature protection OPT[0]: 0 = Not used for cell temperature protection 1 = Used for cell temperature protection
cfetoff_pin_fxn	[1: 0]	0x0	RW	cfetoff_pin_fxn [1:0] 0x0 RW These bits configure which function this pin is used for: 0 = SPI_CS 1 = General Output 2 = CFETOFF 3 = ADC Input or Thermistor

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8 Cell Balancing

8.1 Cell Balancing Overview

The GD30BM2016 supports passive cell balancing by bypassing the current of a selected cell during charging or at rest, using either integrated bypass switches between cells, or external bypass MOS / BJT switches. The device incorporates a voltage-based balancing algorithm which can optionally balance cells autonomously without requiring any interaction with a host processor. Or if preferred, balancing can be entirely controlled manually from a host processor. For autonomous balancing, the device will only balance

non-adjacent cells in use (it does not consider inputs used to measure interconnect as cells in use). To avoid excessive power dissipation within the GD30BM2016, the maximum number of cells allowed to balance simultaneously can be limited by configuration setting. For host-controlled balancing, adjacent as well as non-adjacent cells can be balanced. Host-controlled balancing can be controlled using specific subcommands sent by the host. The device also returns status information regarding how long cells have been balanced through subcommands.

For autonomous balancing, the microprocessor can configure the following parameters:

- Balancing enable states, such as SLEEP balancing, RELAX balancing, CHG balancing, etc.
- Maximum number of balancing cells
- Minimum balancing voltage threshold
- Minimum and maximum cell temperature for balancing
- Maximum internal chip temperature
- Threshold of voltage difference between balancing cell voltage and minimum cell voltage

Table 2 is the **BALANCIN_ CONFIGURATION Register**. **cb_en** is the overall balancing enable switch. It is recommended disable **cb_alert_mask**, since some alert information is not critical and does not need to interrupt balancing, balancing has its own protection mechanism. **cb_no_cmd** determines whether to accept balancing commands from the host. **cb_nosleep/sleep** configures whether balancing is allowed during sleep mode. **cb_rlx** represents the relax state, meaning whether balancing is enabled under the normal condition when the system is neither charging nor discharging. **cb_chg** indicates whether balancing is enabled during charging. If the system enables balancing, the **Power Config: loop_slow** bit also needs to be set to a non-zero value.

Table 2. BALANCIN_ CONFIGURATION Register

Field	Bits	Default	Type	Field Description
opw_en	[7: 7]	0x1	RW	0: open_wire detection function is disabled 1: open_wire detection function is enabled
cb_en	[6: 6]	0x0	RW	0: cell balancing function is disabled 1: cell balancing function is enabled
cb_alert_mask	[5: 5]	0x0	RW	0: ALERT Dont Break cell balance 1: ALERT Will Break cell balance
cb_no_cmd	[4: 4]	0x0	RW	0: Host-controlled balancing commands are accepted 1: Host-controlled balancing commands are ignored
cb_nosleep	[3: 3]	0x0	RW	0: sleeep mode is allowed while cell balancing is active

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Field	Bits	Default	Type	Field Description
				1: sleeep mode is prevented while cell balancing is active
cb_sleep	[2: 2]	0x0	RW	0: Cell balancing is not performed in sleep mode 1: Cell balancing may be performed in sleep mode
cb_rlx	[1: 1]	0x0	RW	0: Cell balancing is not allowed in relax conditions 1: Cell balancing is allowed in relax conditions
cb_chg	[0: 0]	0x0	RW	0: Cell balancing is not allowed while charging 1: Cell balancing is allowed while charging

After the above threshold conditions are met, the internal logic of the device will evaluate its own temperature and the temperature range of the battery pack. If all conditions are satisfied, the device will automatically initiate cell balancing.

The balancing operation works in conjunction with a timer. When the timer times out, the internal logic of the device will recheck all balancing conditions. Balancing can only restart once all conditions are satisfied again, ensuring that the system's balancing process remains stable and safe.

In manual cell balancing mode, the device performs balancing operations based on the cell balancing bits in the register. Once the cell balancing bits are set, the cell balancing process starts and continues to run until the timer times out.

To handle potential abnormalities such as MCU runaway, once the balancing process stops due to a timeout, the microcontroller must reset the balancing bits to restart the cell balancing. Otherwise, the balancing process will remain stopped and cannot continue.

Whether in autonomous or manual balancing mode, the balancing operation starts after the sampling process is completed. This design prevents the balancing process from introducing errors into cell voltage sampling. The internal balancing control adopts a duty-cycle method, meaning that balancing is periodically turned on and off rather than remaining continuously active throughout the entire cycle. The main purpose of this design is to prevent the internal balancing MOSFET from overheating, while also allowing cell voltage measurements to be taken during the intervals between balancing periods.

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9 High-side MOSFET Drivers

9.1 High-side MOSFET Drivers Overview

The GD30BM2016 includes an integrated charge pump and high-side MOSFET drivers for driving CHG and DSG protection N-MOSFETs. In addition, the device supports the use of P-MOSFETs to implement precharge and predischarge functions. By connecting resistors in series in the precharge and predischarge paths, the precharge and predischarge currents can be configured and adjusted to meet different application requirements.

9.2 Charge Pump

The GD30BM2016 integrates an internal charge pump that drives the high-side N-MOSFETs required for charge and discharge control. In practical applications, users can disable the charge pump to reduce power consumption, particularly in low-power sleep mode. Additionally, the device supports low-current discharge operation through N-channel MOSFETs to meet specific low-power requirements.

- In Sleep mode, the charge pump (CP) can be enabled by setting the **CHG_PUMP_CONTROL: sleep_cpen** bit. The driver is activated only after the CP voltage reaches a stable level. The CP output voltage provides two configurable options: 5.5V and 11V.
- By configuring the **CHG_PUMP_CONTROL: sfmode_sleep** bit, the discharge MOSFET can be driven either by the CP or by VBAT (source follower configuration).

9.3 Charge and Discharge

The GD30BM2016 supports both series configuration and parallel configuration of the system's charge and discharge MOSFETs. It can also operate using only the charge MOSFET or only the discharge MOSFET independently.

In the series MOSFET configuration, the charge port and discharge port are shared. In this application scenario, when the device turns on one of the MOSFETs, the body diode protection function will automatically activate once the current exceeds the preset threshold.

If the charge MOSFET is turned off while the discharge MOSFET and pre-discharge MOSFET are turned on, and the detected discharge current amplitude exceeds the programmable threshold (indicating a significant discharge current), the device will automatically turn on the charge MOSFET to prevent current from flowing through the body diode of the charge MOSFET, which could otherwise cause damage. When the discharge current drops below the threshold, if the condition that caused the charge MOSFET to be turned off still exists, the charge MOSFET will be turned off again.

If the discharge MOSFET is turned off while the charge MOSFET and pre-charge MOSFET are turned on, and the detected charge current exceeds the programmable threshold (indicating a significant charging current), the device will automatically turn on the discharge MOSFET to prevent current from flowing through the body diode of the discharge MOSFET, thereby avoiding potential damage. When the charging current drops below the threshold, if the condition that caused the discharge MOSFET to be turned off still exists, the discharge MOSFET will be turned off again.

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9.4 Charge and Discharge N-MOSFETs Control Mode

For the control of charge and discharge MOSFETs, the device supports fully autonomous control, Partially autonomous control, and manual control.

- **fet_init_off** – Configures whether the MOSFET is allowed to turn on when powered up in the initial state;
- **pdsg_en** – Configures whether to enable PDSG before enabling DSG;
- **fet_ctrl_en** – Configures the fully autonomous mode; the device autonomously turns the MOSFET on or off according to the preset protection and recovery conditions. It is recommended to enable this bit so that the Device can independently protect the battery system even if the MCU fails;
- **host_fet_en** – Configures whether the main controller is allowed to control the device. When this bit is set to 1, the device operates in partially autonomous mode, allowing both the device and the external controller to control the MOSFET switching simultaneously.

Table 3. FET_OPTIONS Register

Field	Bits	Default	Type	Field Description
fet_init_off	[5: 5]	0x0	RW	0: Default host FET control state allows FETs to be on 1: Default host FET control state forces FETs off
pdsg_en	[4: 4]	0x0	RW	0: PDSG FET is not used 1: PDSG FET is turned on before DSG
fet_ctrl_en	[3: 3]	0x1	RW	0: FETs will not be turned on 1: FETs are controlled by the device
host_fet_en	[2: 2]	0x1	RW	0: Host FET control commands are ignored 1: Host FET control commands are allowed
sleepchg	[1: 1]	0x0	RW	0: CHG FET is turned off in SLEEP mode 1: CHG FET may be enabled in SLEEP mode
sfet	[0: 0]	0x1	RW	0: Parallel FET mode: Body diode protection is disabled 1: Series FET mode: Body diode protection is enabled

9.4.1 Fully Autonomous Charge and Discharge N-MOSFETs Control

The GD30BM2016 enters the fully autonomous mode by default, which can be used in both Normal and Sleep mode. In the fully autonomous mode, the device autonomously performs the following operations:

- Autonomously detects faults and turns off the MOSFET.
- Autonomously monitors recovery conditions and re-enables the MOSFET when the recovery criteria are met.
- If a secondary protection fault occurs, the device will keep the MOSFET turned off indefinitely to ensure system safety.

9.4.2 Partially Autonomous Charge and Discharge N-MOSFETs Control

The user can configure the device to operate in partially autonomous mode under Normal, Sleep, and DeepSleep modes. In this mode, the device automatically detects faults and shuts down the MOSFETs. If the MCU's command to turn off the MOSFETs has not been cleared, the device will not automatically turn the MOSFETs back on, and the recovery condition must be monitored and controlled by the host controller.

- If a secondary protection (PF) event occurs, the device will autonomously and indefinitely turn off the

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MOSFETs to ensure system safety.

- When a fault event occurs, the device autonomously turns off the corresponding MOSFETs until the host controller decides to re-enable them.

The host controller can also control the MOSFET state through the following methods:

- Use the GPIO of the host controller to control the CFETOFF/DFETOFF pins to turn off the corresponding MOSFETs.
- Use the commands **DSG_PDSG_OFF**, **CHG_PCHG_OFF**, and **ALL_FETS_OFF** to keep the corresponding MOSFETs in the OFF state.
- Use the subcommands **ALL_FETS_OFF** and **ALL_FETS_ON** to individually enable or disable each MOSFET.

In addition, the **FET_ENABLE** subcommand can be used to toggle the original MOSFET ON or OFF state.

9.4.3 Manual Charge and Discharge N-MOSFETs Control

The user can configure the device to operate in manual mode during the Normal, Sleep, and DeepSleep states. In this mode, the main controller is responsible for detecting fault interrupts and manually turning off the MOSFETs, while monitoring the recovery conditions. Once the recovery conditions are met, the main controller manually turns the MOSFETs back on.

When a fault event occurs, the device generates an interrupt through the ALERT pin. After receiving the interrupt, the main controller reads the fault status information and identifies the abnormal condition. At this point, the main controller can turn off the corresponding MOSFETs in the following ways:

- Use the CFETOFF/DFETOFF pins to control and turn off the corresponding MOSFETs.
- Use the commands **DSG_PDSG_OFF**, **CHG_PCHG_OFF**, and **ALL_FETS_OFF** to keep the corresponding MOSFETs turned off until the main controller decides to turn them back on.

9.4.4 Precharge and Predischarge P-MOSFETs Control Mode

The GD30BM2016 supports a precharge mode, which utilizes a high-side P-MOSFET (driven by the PCHG pin) and a series resistor to perform charging, thereby reducing the charging current for undervoltage battery cells. The pre-charge mode continues until the individual cell voltage reaches the preset threshold, at which point it automatically turns off. If the minimum cell voltage in the battery pack is lower than the threshold defined by the register **PRECHARGE_START_VOLTAGE**, the device enters the precharge mode. In Normal state, the voltage check is performed every 100 ms, and in Sleep state, every 5 s. When the battery voltage exceeds the threshold defined by **PRECHARGE_STOP_VOLTAGE**, the device exits the precharge mode. Even if the main controller configures the device for normal charging, the device will prioritize using the precharge MOSFET for charging. If the user wishes to disable this function, the threshold can be set very low to prevent entry into the precharge mode.

In addition, the GD30BM2016 also supports a predischarge mode, which is implemented using a high-side PDSG P-MOSFET (driven by the PDSG pin) and a series resistor. When the load is initially powered on, the predischarge function effectively suppresses inrush current and reduces sparking. If the predischarge mode is enabled, each time the load is connected, the device first activates the predischarge MOSFET to supply power, then switches to turning on the discharge MOSFET and turning off the predischarge MOSFET. The exit conditions for the pre-discharge mode include:

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- The PDSG duration set in the register **PREDISCHARGE_TIMEOUT** expires.
- The voltage difference between LD and STACK is checked every 100 ms to verify whether it falls within the threshold range defined by the register **PREDISCHARGE_STOP_DELTA**.

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10 Primary and Secondary Protection Subsystems

10.1 Protection Overview

The GD30BM2016 integrates both primary and secondary protection functions. The primary protection is mainly used to control the charge and discharge MOSFETs, while the secondary protection is designed to control an external chemical fuse.

Users can configure protection thresholds, delay times, and recovery thresholds to monitor parameters such as voltage, current, and temperature. When the preset threshold conditions are met, the device automatically triggers either primary or secondary protection according to the corresponding protection enable settings.

For detailed register configuration methods, please refer to the protection settings section in the relevant register documentation to ensure correct implementation of the protection functions and overall system safety.

10.2 Primary Protection Subsystems

10.2.1 Primary Voltage Protection

The GD30BM2016 features comparator-based hardware protection, supporting CELL voltage overvoltage protection (COV), with configurable overvoltage threshold, protection delay after overvoltage, and overvoltage recovery hysteresis. It also supports CELL voltage undervoltage protection (CUV), allowing configuration of the undervoltage threshold, protection delay after undervoltage, and undervoltage recovery hysteresis.

In addition to COV, if the system allows a certain number of overvoltage events to reduce false overvoltage triggers, the device provides a COVL protection mechanism that accumulates the number of overvoltage faults or automatically decrements the count when the overvoltage condition is cleared, thereby improving the accuracy of overvoltage fault detection.

Table 4. Primary Voltage Protection Related Register Settings

Name	Enable	Alarm Mask	Protection Threshold	ALERT Trigger	Action	Delay	FAULT Trigger	Action	Recovery Threshold	Recovery Time
CUV	Default 0: Disabled	Y	CUV_THRESHOLD	Safety Alert A [2] =1	ALERT Pin active	CUV_DELAY	Safety Status A [2] =1	DSG OFF	CUV_RECOVERY_HYSTERESIS	No delay
COV	Default 1: Enabled	Y	COV_THRESHOLD	Safety Alert A [3] =1		COV_DELAY	Safety Status A [3] =1	CHG OFF/COVL+1	COV_RECOVERY_HYSTERESIS	No delay
COVL	Default 0: Disabled	Y	COVL_LATCH_LIMIT	Safety Alert C [4] =1		COVL_COUNTER_DEC_DELAY	Safety Status C [4] =1	CHG OFF		COVL_RECOVERY_TIME

10.2.2 Primary Current Protection

The GD30BM2016 adopts a comparator-based hardware protection mechanism, supporting battery pack charge overcurrent protection (OCC), discharge overcurrent protection 1 (OCD1), discharge overcurrent protection 2

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(OCD2), and short-circuit protection (SCD). The device supports battery pack discharge overcurrent protection 3 (OCD3) through a digital comparator based on ADC current measurement values.

In addition, the device includes OCDL and SCDL protection mechanisms. If the system permits a certain number of overcurrent or short-circuit events to reduce false trigger probability, OCDL and SCDL can be used to accumulate fault counts. When the fault condition disappears and the duration exceeds the delay threshold, the device automatically decrements the count, thereby reducing the likelihood of unnecessary short-circuit protection activation.

Table 5 Primary Current Protection Related Register Settings

Name	Enable	Alarm Mask	Protection Threshold	ALERT Trigger	Action	Delay	FAULT Trigger	Action	Recovery Threshold	Recovery Time
OCC	Default 0: Disabled	Y	OCC_THRESHOLD	Safety Alert A [4] = 1	ALERT Pin active	OCC_DELAY	Safety Status A [4] = 1	CHG OFF	OCC_RECOVERY_THRESHOLD	RECOVERY_TIME
OCD1	Default 0: Disabled	Y	OCD1_THRESHOLD	Safety Alert A [5] = 1		OCD1_DELAY	Safety Status A [5] =1	DSG OFF/OCDL+1	OCD_RECOVERY_THRESHOLD	
OCD2	Default 0: Disabled	Y	OCD2_THRESHOLD	Safety Alert A [6] = 1		OCD2_DELAY	Safety Status A [6] =1			
OCD3	Default 0: Disabled	Y	OCD3_THRESHOLD	Safety Alert C [7] = 1		OCD3_DELAY	Safety Status C [7] =1			
OCDL	Default 0: Disabled	Y	OCDL_LATCH_LIMIT	Safety Alert C [5] = 1		OCDL_COUNTER_DEC_DELAY	Safety Status C [5] =1	DSG OFF	OCDL_RECOVERY_THRESHOLD	OCDL_RECOVERY_TIME
SCD	Default 1: Enabled	Y	SCD_THRESHOLD	Safety Alert A [7] = 1		SCD_DELAY	Safety Status A [7] =1	DSG OFF/ SCDL+1	SCDL_RECOVERY_THRESHOLD	SCD_RECOVERY_TIME
SCDL	Default 0: Disabled	Y	SCDL_LATCH_LIMIT	Safety Alert C [6] = 1		SCDL_COUNTER_DEC_DELAY	Safety Status C [6] =1	DSG OFF	SCDL_RECOVERY_THRESHOLD	SCDL_RECOVERY_TIME

10.2.3 Primary Temperature Protection

The GD30BM2016 employs a digital comparator based on ADC temperature measurement values to support overtemperature and undertemperature protection for battery cells, charge/discharge MOSFETs, and the device itself.

Users can independently configure the protection threshold, delay time, and recovery threshold for cell charge overtemperature (OTC), cell charge undertemperature (UTC), cell discharge overtemperature (OTD), cell discharge undertemperature (UTD), MOSFET overtemperature (OTF), and chip overtemperature (OTINT) to meet the safety requirements of various applications.

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Table 6. Primary Temperature Protection Related Register Settings

Name	Enable	Alarm Mask	Protection Threshold	ALERT Trigger	Action	Delay	FAULT Trigger	Action	Recovery Threshold	Recovery Time
OTC	Default 0: Disabled	Y	OTC_THRESHOLD	Safety Alert B [4] = 1	ALERT Pin active	OTC_DELAY	Safety Status B [4] =1	CHG OFF	OTC_RECOVERY_THRESHOLD	RECOVERY_TIME
OTD	Default 0: Disabled	Y	OTD_THRESHOLD	Safety Alert B [5] = 1		OTD_DELAY	Safety Status B [5] =1	DSG OFF	OTD_RECOVERY_THRESHOLD	
OTF	Default 0: Disabled	Y	OTF_THRESHOLD	Safety Alert B [7] = 1		OTF_DELAY	Safety Status B [7] =1	DSG/CHG/BOTH OFF	OTF_RECOVERY_THRESHOLD	
OTINT	Default 0: Disabled	Y	OTINT_THRESHOLD	Safety Alert B [6] = 1		OTINT_DELAY	Safety Status B [6] =1	DSG/CHG/BOTH OFF	OTINT_DELAY	
UTC	Default 0: Disabled	Y	UTC_THRESHOLD	Safety Alert B [0] = 1		UTC_DELAY	Safety Status B [0] =1	CHG OFF	UTC_RECOVERY_THRESHOLD	
UTD	Default 0: Disabled	Y	UTD_THRESHOLD	Safety Alert B [1] = 1		UTD_DELAY	Safety Status B [1] =1	DSG OFF	UTD_RECOVERY_THRESHOLD	
UTINT	Default 0: Disabled	Y	UTINT_THRESHOLD	Safety Alert B [2] = 1		UTINT_DELAY	Safety Status B [2] =1	DSG/CHG/BOTH OFF	UTINT_RECOVERY_THRESHOLD	

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10.3 Secondary Protection Function

10.3.1 Secondary Protection Function

The secondary protection defines several abnormal conditions. When any of these conditions are detected, the device activates secondary protection, blowing the chemical fuse to permanently disable the PACK's charge and discharge paths.

By default, the detection of these abnormal conditions is disabled. If users need to enable a specific abnormal condition detection, they can set the corresponding bit to enable it. Once the detection is enabled, the alert MASK is automatically activated. When an abnormal condition occurs, the device generates an ALTER interrupt and logs the status, which can be read via the direct command **PF_ALERTX** to obtain the ALERT information.

If the fault persists beyond the delay time configured by the digital filter, the device triggers an error, blows the external chemical fuse, permanently disables the charge and discharge paths, and logs the PF status. The FAULT information can be read via the direct command **PF_STATUSX.**

Table 7. Secondary Protection Related Register Settings

Name	Enable	Alarm Mask	Protection Threshold	ALERT Trigger	Action	Delay	FAULT Trigger	Action	Save PF Status
CUDEP	Default 0: Disabled	Y	CUDEP_THRESHOLD	PF Alert A [7] = 1	ALERTPin active	CUDEP_DELAY	PF Status A [7] = 1	Blowing the chemical fuse	Store PF status
SOTF	Default 0: Disabled	Y	SOTF_THRESHOLD	PF Alert A [6] = 1		SOTF_DELAY	PF Status A [6] = 1		
SOT	Default 0: Disabled	Y	SOT_THRESHOLD	PF Alert A [4] = 1		SOT_DELAY	PF Status A [4] = 1		
SOCD	Default 0: Disabled	Y	SOCD_THRESHOLD	PF Alert A [3] = 1		SOCD_DELAY	PF Status A [3] = 1		
SOCC	Default 0: Disabled	Y	SOCC_THRESHOLD	PF Alert A [2] = 1		SOCC_DELAY	PF Status A [2] = 1		
SOV	Default 0: Disabled	Y	SOV_THRESHOLD	PF Alert A [1] = 1		SOV_DELAY	PF Status A [1] = 1		
SUV	Default 0: Disabled	Y	SUV_THRESHOLD	PF Alert A [0] = 1		SUV_DELAY	PF Status A [0] = 1		
VIMA	Default 0: Disabled	Y	VIMA_THRESHOLD	PF Alert B [4] = 1		VIMA_DELAY	PF Status B [4] = 1		
VIMR	Default 0: Disabled	Y	VIMR_THRESHOLD	PF Alert B [3] = 1		VIMR_DELAY	PF Status B [3] = 1		
2LVL	Default 0: Disabled	Y		PF Alert B [2] = 1		DUALLVL_DELAY	PF Status B [2] = 1		
DFETF	Default 0: Disabled	Y	DFETF_OFF_THRESHOLD	PF Alert B [1] = 1		DFETF_OFF_DELAY	PF Status B [1] = 1		
CFETF	Default 0: Disabled	Y	CFETF_OFF_THRESHOLD	PF Alert B [0] = 1		CFETF_OFF_DELAY	PF Status B [0] = 1		
HWMX	Default 0:	Y		PF Alert C [6]		HWMX_DELA	PF Status C		

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Name	Enable	Alarm Mask	Protection Threshold	ALERT Trigger	Action	Delay	FAULT Trigger	Action	Save PF Status
	Disabled			= 1		Y	[6] = 1		
VSSF	Default 0: Disabled	Y	VSSF_FAIL_THRESHOLD	PF Alert C [5] = 1		VSSF_DELAY	PF Status C [5] = 1		
LFOF	Default 0: Disabled	Y		PF Alert C [3] = 1		LFOF_DELAY	PF Status C [3] = 1		

The Table 7 lists the secondary protection items along with their corresponding alarm MASKs, protection thresholds, and delay times. Each parameter has a corresponding register setting. Please refer to the register documentation for details.

Table 8. Functional description of secondary protection

Name	Functional Description
CUDEP	Copper Deposition Permanent Fail
SOTF	Safety Overtemperature FET Permanent Fail
SOT	Safety Overtemperature Permanent Fail
SOCD	Safety Overcurrent in Discharge Permanent Fail
SOCC	Safety Overcurrent in Charge Permanent Fail
SOV	Safety Cell Overvoltage Permanent Fail
SUV	Safety Cell Undervoltage Permanent Fail
VIMA	Voltage Imbalance Active Permanent Fail
VIMR	Voltage Imbalance at Rest Permanent Fail
2LVL	Second Level Protector Permanent Fail
DFETF	Discharge FET Permanent Fail
CFETF	Charge FET Permanent Fail
HWMX	Internal Stuck Hardware Mux Permanent Fail
VSSF	Internal VSS Measurement Permanent Fail
LFOF	Internal LFO Permanent Fail

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11 Linear regulator power supply

11.1 Linear regulator power supply Overview

The GD30BM2016 contains an integrated 1.8V LDO (REG18) that provides a regulated 1.8 V supply voltage for the device's internal circuitry and digital logic. This regulator uses an external capacitor connected to the REG18 pin, and it should only be used for internal circuitry.

The device also integrates two separately programmable LDOs (REG1 and REG2) for external circuitry, such as a host processor or external transceiver circuitry, which can be programmed to independent output voltages. The REG1 and REG2 LDOs take their input from the REGIN pin, with this voltage either provided externally or generated by an integrated preregulator, referred to as REG0. The REG1 and REG2 LDOs can provide an output current of up to 45 mA each.

11.2 Preregulator Control

The input voltage of the REG1 and REG2 LDOs is supplied from the REGIN pin, which can be generated by an integrated voltage preregulator, referred to as REG0. REG0 drives the base of an external NPN BJT through the BREG pin to provide a 5.5 V voltage at the REGIN pin. The REGIN pin voltage can also be supplied externally, for example, by an independent DC/DC converter.

When using the REG0, appropriate transistors and series resistors should be selected according to the battery voltage and load current to meet thermal dissipation requirements.

11.3 REG1 and REG2 LDO Controls

The REG1 and REG2 LDOs in the device are for customer use, and their output voltages can be programmed independently to 1.8 V, 2.5 V, 3.0 V, 3.3 V, or 5.0 V. Users can independently configure whether REG1 and REG2 LDOs are enabled in Sleep or DeepSleep mode to reduce the overall system power consumption during standby.

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12 Multifunction Pin

12.1 Multifunction Pins Overview

To facilitate flexible use of digital I/O and internal ADC resources, the GD30BM2016 is designed with multiple multifunction pins, including TS1, TS2, TS3, CFETOFF, DFETOFF, ALERT, SPI_MOSI, SCL, SDA, DCHG, and DDSG. Each pin can be independently configured for different functions.

Table 9 describes the configurable functions of each pin, which can be set through the lower two bits of the multifunction pin configuration register.

- ALERT Signal: Used for system interrupt functions, including interrupts triggered by primary protection warning, primary protection fault, and secondary protection warning. It can also be configured as a digital GPIO, ADC input, or thermistor input.
- DDSG/DCHG Signals: Serve as shadow signals for high-side drive signals, indicating the enable status of charge (CHG) and discharge (DSG) signals. In low-side drive applications, they can be used as input signals for low-side driver chips or circuits. They can also function as digital GPIOs, ADC inputs, or thermistor inputs.
- CFETOFF/DFETOFF: Can be used by the main controller via the I/O interface to directly control the operation of charge and discharge MOSFETs in hardware. They can also be configured as digital GPIOs, ADC inputs, or thermistor inputs. CFETOFF can also be configured as the SPI chip select signal pin.
- SPI_MOSI: Data signal in SPI mode, used for address setting in I²C mode.They can also function as digital GPIOs, ADC inputs, or thermistor inputs.

Table 9. Multifunction Pins Description

Pin Name	Type[1]	Function Description
TS1	I/OA	ts1_opt [3:2] configure which function this pin is used for: ts1_opt [2] = 1, Thermistor input ts1_opt [3] = 1, ADC input
TS2	I/OA	ts2_opt [3:2] configure which function this pin is used for: ts2_opt [2] = 1, Thermistor input ts2_opt [3] = 1, ADC input
TS3	I/OA	ts3_opt [3:2] configure which function this pin is used for: ts3_opt [2] = 1, Thermistor input ts3_opt [3] = 1, ADC input
ALERT	OD, I/OA	alert_pin_fxn [1:0] configure which function this pin is used for: 0 = Unused 1 = General Output 2 = ALERT 3 = ADC Input or Thermistor (alert_opt [2] = 1, Thermistor input; alert_opt [3] = 1, ADC input)
SPI_MOSI	I/OD, I/OA	spi_mosi_pin_fxn [1:0] configure which function this pin is used for: 0 = SPI_MOSI or I2C address pin 1 = General Output

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Pin Name	Type[1]	Function Description
		2 = Unused Not used 3 = ADC Input or Thermistor (spi_mosi_opt [2] = 1, Thermistor input; spi_mosi _opt [3] = 1, ADC input)
CFETOFF	I/OD, I/OA	cfetoff_pin_fxn [1:0] configure which function this pin is used for: 0 = SPI_CS 1 = General Output 2 = CFETOFF 3 = ADC Input or Thermistor (cfetoff_opt [2] = 1, Thermistor input; cfetoff_opt [3] = 1, ADC input)
DFETOFF	I/OD, I/OA	dfetoff_pin_fxn [1:0] configure which function this pin is used for. 0 = DFETOFF 1 = General Purpose Output 2 = BOTHOFF 3 = ADC Input or Thermistor (dfetoff_opt [2] = 1, Thermistor input; dfetoff_opt [3] = 1, ADC input)
DCHG	OD, I/OA	dchg_pin_fxn[1:0] configure which function this pin is used for: 0 = Reserved. 1 = General Output. 2 = DCHG. 3 = ADC Input or Thermistor (dchg_opt [2] = 1, Thermistor input; dchg_opt [3] = 1, ADC input)
DDSG	OD, I/OA	ddsg_pin_fxn [1:0] configure which function this pin is used for: 0 = Reserved 1 = General Output 2 = DDSG 3 = ADC Input or Thermistor (ddsg_opt [2] = 1, Thermistor input; ddsg_opt [3] = 1, ADC input)
SCL	I/OD	scl_pin_fxn [1:0] configure which function this pin is used for: 0 = SCL or SPI_SCLK 1 = Not used 2 = Not used 3 = Not used
SDA	I/OD	sda_pin_fxn [1:0] configure which function this pin is used for: 0 = SDA or SPI_MISO 1 = Not used 2 = Not used 3 = Not used

1. IA = Analog Input; OA = Analog Output; ID = Digital Input; OD = Digital Output.

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13 Operating Mode

13.1 Operating Mode Overview

The GD30BM2016 supports four operating modes, including Normal, Sleep, DeepSleep and ShutDown mode, to achieve optimized features and power consumption. The device can transition between modes either autonomously or under the control of a host processor. In these modes, the internal functional modules of the device can be flexibly configured to be enabled or disabled to meet different power consumption and functional requirements.

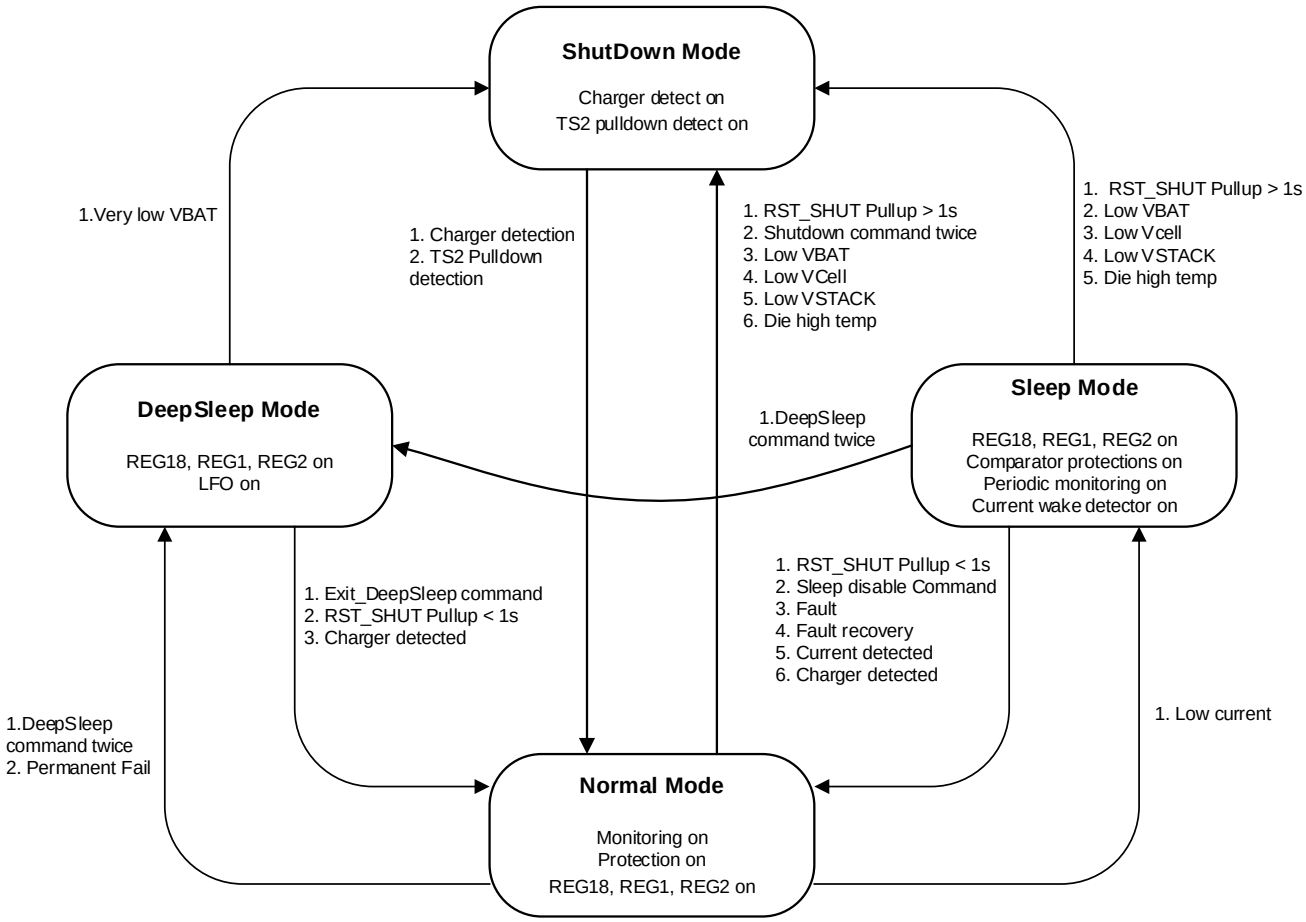


Figure 2. Device Operating Modes

13.2 Normal Mode

In Normal mode, the device performs periodic measurements of voltage, current and temperature, with REG18 and LFO operating. Based on the device configuration settings, full battery protection functions are active. This mode represents the highest performance mode of the device.

The device enters Sleep mode from Normal mode under the following conditions:

- The CC1 current is lower than the threshold set in the **SLEEP_CURRENT Register**.

The device enters DeepSleep mode from Normal mode under the following conditions:

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- Send the **DeepSleep** command twice consecutively within 4 seconds.
- In the event of secondary protection with the chemical fuse blown, it is necessary to wait for the fuse blow time.

The device enters ShutDown mode from Normal mode under the following conditions:

- After the RST_SHUT signal remains high for more than 1 second and a period of delay is applied according to the threshold configured in the **Shutdown Command Delay Register**.
- After two indirect **SHUTDOWN** commands are received consecutively within 4 seconds, and a period of delay is applied according to the threshold configured in the **Shutdown Command Delay Register**.
- After the on-chip over-temperature reaches the temperature threshold configured in the **Shutdown Temp Register** and persists for the duration configured in the **Shutdown Temperature Delay Register**.
- After the cell voltages drop to the threshold set in the **Shutdown Cell Voltage Register** and remain below it for the duration configured in the **Low Shutdown Delay Register**.
- After the Stack voltage drops to the threshold set in the **Shutdown Stack Voltage Register** and remains below it for the duration configured in the **Low Shutdown Delay Register**.
- When the VBAT voltage falls below V_{POR} - V_{POR_HYS}, an analog POR reset is triggered, causing the device to enter ShutDown mode.

13.3 Sleep Mode

In Sleep mode, the device supports periodic monitoring of voltage, temperature, and current, as well as comparator protection functions, while keeping internal modules such as REG18 and LFO enabled. This mode is a reduced-functionality state that can be optionally used when the system load current is very low or no charging is in progress, in order to reduce power consumption, while still maintaining voltage at the battery pack terminals to keep the system operational.

The device enters Normal mode from Sleep mode under the following conditions:

- When the measured current exceeds the threshold configured in the **Wake Comparator Current Register,** the device immediately exits Sleep mode and quickly turns on the FETs (if the CHG FET is off or the DSG FET is in source-follower mode).
- The RST_SHUT signal stays high for more than 200 ms but less than 1 s.
- When the PACK voltage exceeds the STACK voltage by more than the configured threshold (set via the **Sleep Charger Packtos Delta Register**) and the STACK voltage is below the configured threshold (set via the **Sleep Charger Voltage Threshold Register**), indicating a charger connection condition.
- The **Sleep disable** command is received.
- When a fault condition occurs.
- Upon fault recovery.
- When a secondary protection PF event occurs.

The device enters DeepSleep mode from Sleep mode under the following conditions:

- Send the DeepSleep command twice consecutively within 4 seconds.

The device enters ShutDown mode from Sleep mode under the following conditions:

- The RST_SHUT signal remains high for more than 1 second, and after a delay defined by the **Shutdown Command Delay Register** threshold.

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- The on-chip over-temperature reaches the threshold set in the **Shutdown Temp Register** and persists longer than the duration configured in the **Shutdown Temperature Delay Register**.
- After the cell voltages drop to the threshold set in the **Shutdown Cell Voltage Register** and remain below it for the duration configured in the **Low Shutdown Delay Register**.
- After the Stack voltage drops to the threshold set in the **Shutdown Stack Voltage Register** and remains below it for the duration configured in the **Low Shutdown Delay Register**.
- When the VBAT voltage falls below V_{POR} - V_{POR_HYS}, an analog POR reset is triggered, causing the device to enter ShutDown mode.

13.4 DeepSleep Mode

In DeepSleep mode, the protection FETs are all disabled, so no voltage is provided at the battery pack terminals. All protections are disabled, and all voltage, current, and temperature measurements are disabled. This mode is a low-power mode, users can configure the registers to keep REG1/2 operating in DeepSleep mode **(Power_Config: dpslp_ldo** = 1), and can also configure the LFO to remain active in DeepSleep mode (**Power_Config: dpslp_lfo** = 1).

The device enters Normal mode from DeepSleep mode under the following conditions:

- Charging connection detected.
- The RST_SHUT signal remains at a high level for more than 200 ms and less than 1 s.
- Send the Normal command.

The device enters ShutDown mode from DeepSleep mode under the following conditions:

- When the VBAT voltage falls below V_{POR} - V_{POR_HYS}, an analog POR reset is triggered, causing the device to enter ShutDown mode.

13.5 ShutDown Mode

In ShutDown mode, the device only retains the charge detection function and the TS2 pin wake-up function. In this mode, the device loses all register state information, the internal logic is powered down, the protection FETs are all disabled, so no voltage is provided at the battery pack terminals. All protections are disabled, all voltage, current, and temperature measurements are disabled, and no communications are supported. ShutDown mode is the lowest power mode, which can be used for shipping or long-term storage.

When the device exits ShutDown Mode, it will boot and read parameters stored in OTP (if that has been written). If the OTP has not been written, the device will power up with default settings, and then settings can be changed by the host writing device registers.

The device enters Normal mode from ShutDown mode under the following conditions:

- Charger connection detected.
- TS2 pin pulled low detected.

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14 Product Configuration and OTP

14.1 Commands and Subcommands

GD30BM2016 supports two operation modes: direct commands and subcommands. Direct commands are accessed through a 7-bit command address sent by the host controller and can be used to trigger an action, write data to the device, or read data from it. Subcommands are indirectly accessed through the 7-bit command address space and provide block data transfer functionality. Subcommands are classified into two types: data subcommands and non-data subcommands.

14.2 Configuring the Device Using OTP or Registers

The GD30BM2016 integrates internal registers whose values are stored in RAM and can be accessed for read and write operations via the I2C and SPI interfaces. Upon initial power-up, the device automatically loads default settings from the OTP into the registers, which are used during device operation.

The recommended OTP usage procedure is as follows: during production, customers should write the initialization configuration into the OTP so that the device uses these fixed settings each time it powers up. If the OTP has not been programmed by the user, the device will use factory default configurations after power-up. The host controller can also initialize the registers after power-up to override the values loaded from the OTP; however, the registers must be reinitialized after every power cycle.

When the device is in Normal, Sleep, or DeepSleep mode, the register values remain unchanged. If the device enters ShutDown mode, all register memory will be cleared, and upon the next power-up, the device will restore default parameters or load the OTP configuration if programmed.

To program the OTP, the user must apply a voltage greater than 12V to the BAT pin, connect a 2.2μF capacitor between pin 19 (NC) and ground, and ensure the device is in FULLACCESS mode.

14.3 Register Configuration Procedure

The device access mechanism is divided into three modes:

- SEALED mode: This is the most secure mode, allowing all direct commands to be accessed in simulation, while some indirect subcommands can also be accessed.
- UNSEALED mode: Some additional subcommands can be accessed in this mode.
- FULLACCESS mode: Used to modify register settings. However, since register modifications may cause measurement and protection abnormalities, it is necessary to enter this mode to perform related operations, and then switch to CFGUPDATE mode for register configuration.

The overall operation procedure is as follows:

- Before entering the FULLACCESS mode, the device needs to switch from SEALED mode to UNSEALED mode through an indirect subcommand, and then from UNSEALED mode to FULLACCESS mode.
- Use the indirect subcommand **SET_CFGUPDATE** to configure the related registers.
- After the configuration is completed, use the indirect subcommand **EXIT_CFGUPDATE**.
- After completing all operations, the device needs to return to SEAL mode.

Detailed process steps:

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- Indirect subcommand with data (**SECURITY_ACCESS**, **unseal_key_step1**, **unseal_key_step2**).
- Indirect subcommand with data (**SECURITY_ACCESS**, **full_acc_key_step1**, **full_acc_key_step2**).
- Execute indirect subcommand (**SET_CFGUPDATE**).
- Complete the register configuration, perform device initialization and related settings.
- Indirect subcommand (**EXIT_CFGUPDATE**).
- Indirect subcommand (**SEAL**).

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15 Serial Communication Interface

15.1 Serial Communication Interface Overview

GD30BM2016 integrates two types of serial communication interfaces:

- I2C bus: supports 100 kHz and 400 kHz rate modes, with optional CRC check.
- SPI bus: supports efficient transmission, with optional CRC check.

Different device versions can support different communication modes and CRC check functions. Users can program the integrated OTP on the production line to configure the required communication speed and protocol for device operation upon power-up.

15.2 I2C Communication

15.2.1 I2C Communication Overview

The I2C serial interface in GD30BM2016 supports a maximum communication rate of 400 kHz and optionally enables the CRC check function. If the OTP is not programmed, the device will operate in 400 kHz I2C mode by default at the first power-up, while other device versions may operate in different communication modes at the first power-up. Users can program the OTP on the production line to set the desired I2C mode for device operation upon power-up.

In addition, the host can modify the I2C speed setting in CONFIG_UPDATE mode, and the new speed setting will take effect upon exiting CONFIG_UPDATE mode. The master controller can also instantly switch the communication interface to I2C through the subcommand **SWAP_TO_I2C**.

The I2C device address (an 8-bit value including the device address and the R/W bit) is set by default to 0x20 (write) and 0x21 (read).

The communication interface supports a programmable timeout function, which is only applicable when the bus operates in 100 kHz or 400 kHz mode. When the device is configured to 100 kHz mode and this function is enabled, the device will reset the internal logic of the communication interface if any of the following conditions occur:

- If a clock is detected low longer than a $t_{TIMEOUT}$ of 25 ms to 35 ms.
- If the cumulative clock low responder extend time exceeds ≈25 ms.
- if the cumulative clock low controller extend time exceeds 10 ms.

If the timeouts are enabled with the device set to 400kHz mode, then the device will reset the communications interface logic if a clock is detected low longer than $t_{TIMEOUT}$ of 5 ms to 20 ms.

The bus also includes a long-term timeout if the SCL pin is detected low for more than 2 seconds, which applies whether or not the timeouts above are enabled.

15.2.2 I2C Direct Command

I2C direct command format：

- S is the start bit.
- RS is a restart.

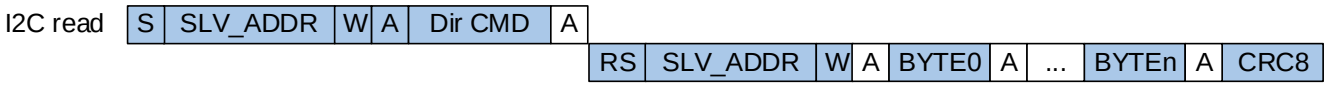
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- SLV_ADDR is the device slave address.
- W is wait.
- N is the last data bit.
- A is the ACK response bit.
- DirCMD is the Direct Command ID.
- BYTEn is the valid data byte.
- P is the stop bit.

I2C Write Command：

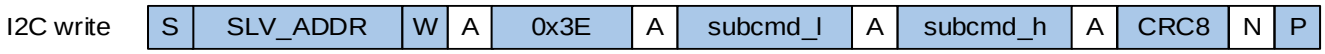


I2C Read Command：



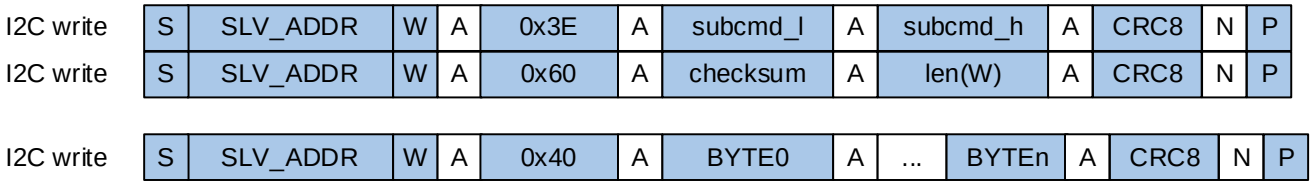
15.2.3 I2C Subcommand without data

I2C Write Subcommand without Data：

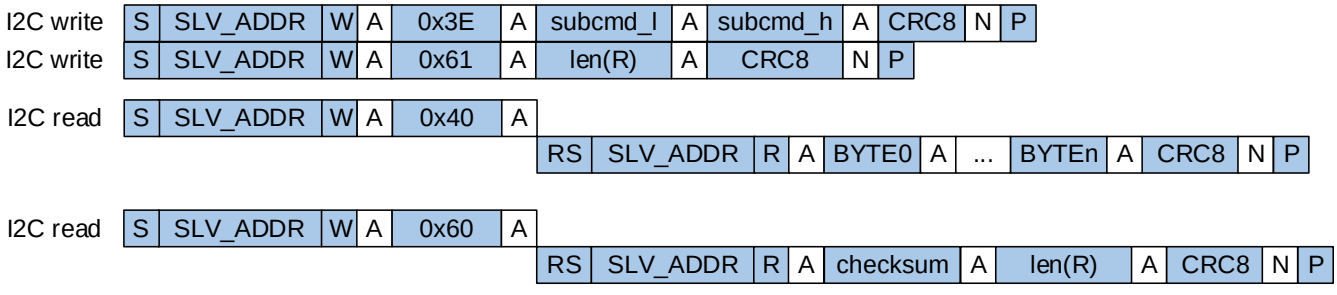


15.2.4 I2C Subcommand with data

I2C Write Subcommand with Data：



I2C Read Subcommand with Data：



15.3 SPI Communication

15.3.1 SPI Communication Overview

The SPI interface in the GD30BM2016 operates as a responder-only interface with an optional CRC check. If the OTP has not been programmed, the device initially powers up by default in 400 kHz I2C mode, while other device versions will initially power up by default in SPI mode with CRC enabled, as described in the Device Comparison

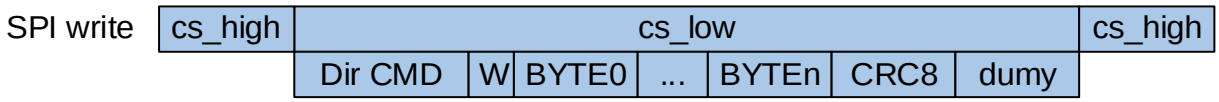
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Table. The OTP setting to select SPI mode can be programmed into the GD30BM2016 on the manufacturing line, then when the device powers up, it automatically enters SPI mode. The host can also change the serial communication setting while in CONFIG_UPDATE mode, although the device will not immediately change communication mode upon exit of CONFIG_UPDATE mode to avoid losing communications during evaluation or production. The host can reset the device or write the **SWAP_TO_SPI** subcommand to change the communications interface to SPI immediately.

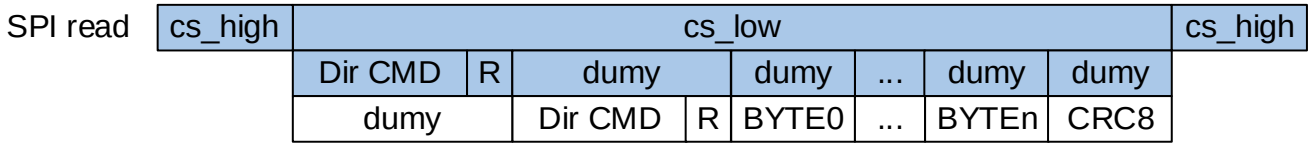
15.3.2 SPI Direct Command

In the SPI command definitions, cs_high represents pulling the SPI CS pin high, cs_low represents pulling the SPI CS pin low; dumy represents a dummy command (0x00); DirCMD is the Direct Command ID; BYTEn refers to a data byte.

SPI Write Command：

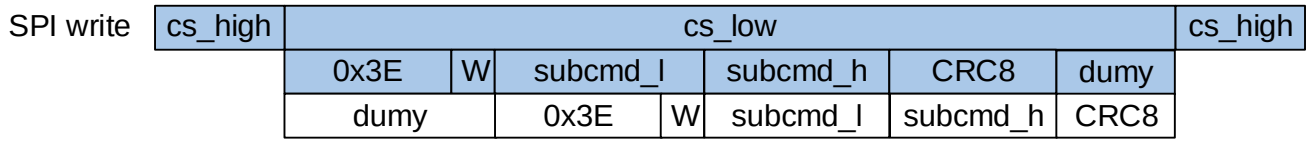


SPI Read Command：



15.3.3 SPI Subcommand without data

SPI Write Subcommand without Data：



15.3.4 SPI Subcommand with data

SPI Write Subcommand with Data：

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SPI write

cs_high	cs_low						cs_high
	0x3E	W	subcmd_l	subcmd_h	CRC8	dumy	
	dumy		0x3E	W	subcmd_l	subcmd_h	CRC8

SPI write

cs_high	cs_low						cs_high
	0x60	W	checksum	len(W)	CRC8	dumy	
	dumy		0x60	W	checksum	len(W)	CRC8

SPI write

cs_high	cs_low							cs_high
	0x40	W	BYTE0	...	BYTEn	CRC8	dumy	
	dumy		0x40	W	BYTE0	...	BYTEn	CRC8

SPI Read Subcommand with Data：

SPI write

cs_high	cs_low						cs_high
	0x3E	W	subcmd_l	subcmd_h	CRC8	dumy	
	dumy		0x3E	W	subcmd_l	subcmd_h	CRC8

SPI write

cs_high	cs_low					cs_high
	0x61	W	len(R)		CRC8	dumy
	dumy		0x61	W	len(R)	CRC8

SPI read

cs_high	cs_low							cs_high
	0x40	R	dumy	dumy	...	dumy	dumy	
	dumy		0x40	R	BYTE0	...	BYTEn	CRC8

SPI read

cs_high	cs_low						cs_high
	0x60	R	dumy	dumy	dumy	dumy	
	dumy		0x60	R	checksum	len(R)	CRC8

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16 Application and Implementation

16.1 Application Information

The GD30BM2016 can be used with 3 to 16 cells battery packs, supporting a top-of-stack voltage ranging from 8 V up to 80 V. The device can accurately monitor the voltage of each cell, temperature, the battery pack current, and the MOSFET temperature, preventing lithium-ion and $LiFePO_4$ battery packs from cell overvoltage, undervoltage, overheating, charge/discharge overcurrent, and discharge short-circuit conditions. It adopts a high-side N- MOSFET architecture with strong drive switching capability.

16.2 Typical Applications

A simplified application schematic for a 16 cells battery pack is shown in Figure 3, using the GD30BM2016 together with an external secondary protector, a host microcontroller, and a communications transceiver. This configuration uses CHG and DSG N-MOSFETs in series, together with high-side P-MOSFET devices used to implement precharge and predischarge functionality. Several points to consider in an implementation are included below:

- The external NPN BJT used for the REGIN preregulator can be configured with its collector routed either to the cell battery stack or the middle of the protection MOSFETs. A diode and a resistor is recommended in the collector circuit of the external NPN BJT, the diode is used to prevent reverse current from flowing from the BREG pin through the BJT base to the collector in the event of a pack short circuit, the resistor is used to share part of the power dissipation during normal operation to prevent the BJT from overheating.
- A series diode is recommended at the BAT pin, together with a capacitor from the pin to VSS. These components allow the device to continue operating for a short time when a pack short circuit occurs, which may cause the PACK+ and top-of-stack voltages to drop to approximately 0 V. In this case, the diode prevents the BAT pin from being pulled low with the stack, and the device will continue to operate, drawing current from the capacitor. Generally operation is only required for a short time, until the device detects the short circuit event and disables the DSG FET.
- The PACK and LD pins should be connected to the top of the battery pack through a 10kΩ series resistor to reduce transient voltage impact on the IC and protect the device from surge effects.
- It is recommended to connect 100 Ω resistors in series with the SRP and SRN pins, and to place a 100 nF capacitor in parallel with an optional 100 pF differential filter capacitor between the two pins for signal filtering. The routing from these components and the sense resistor to the pins should be kept as short and fully symmetrical as possible, and all components are recommended to be placed on the same side of the PCB as the device. To further suppress noise, optional 0.1 µF filter capacitors can be added between each sense input pin and VSS.
- It is recommended to install a reverse-protection diode between PACK+ and PACK- to prevent transient voltage drops caused by load reverse connection or the operation of large inductive loads. The reverse-protection diode can provide freewheeling protection for the inductive load during transient conditions.
- Add resistors between CHG, DSG, and the gate of the N-channel MOSFET to adjust the turn-off waveform and speed. Excessively large resistance may cause the MOSFET to turn off too slowly, leading to overheating, excessively small resistance may result in large surge impacts on the battery terminal, causing the MOSFET's drain-source voltage to become too high. In addition, a Zener diode should be added between the gate and source of the N-channel MOSFET to prevent potential high-voltage damage to the gate. During short-circuit events or overcharge conditions, the drain-source voltage may exceed the MOSFET's rated

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voltage; adding a high-voltage TVS diode can prevent drain-source breakdown of the MOSFET.

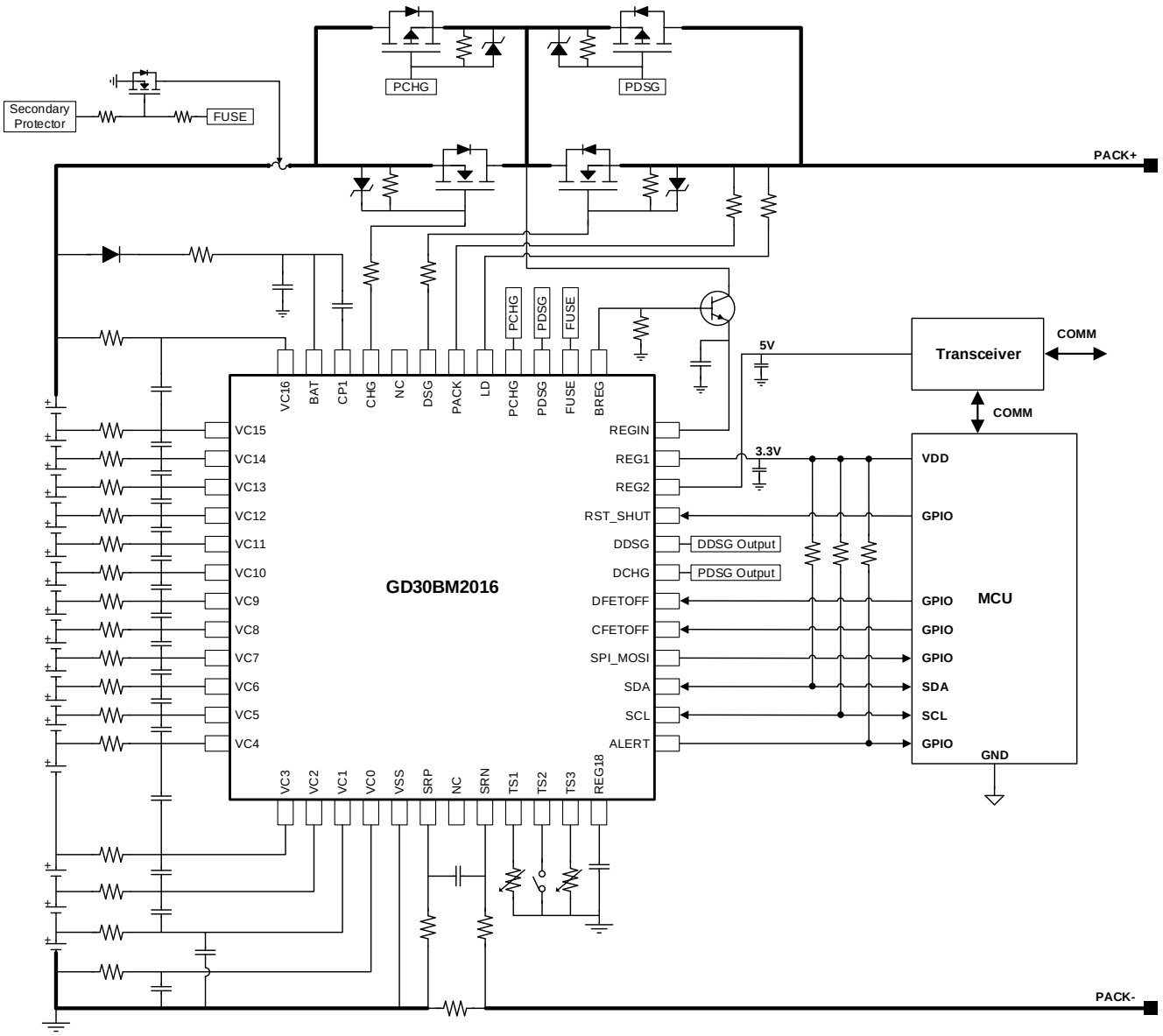


Figure 3. 16 Cells Typical Implementation (Simplified Schematic)

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16.3 Unused Pins

In certain applications, some device pins may not be required. In such cases, Table 10 describes the handling method for each pin.

Table 10. Unused pin handling method

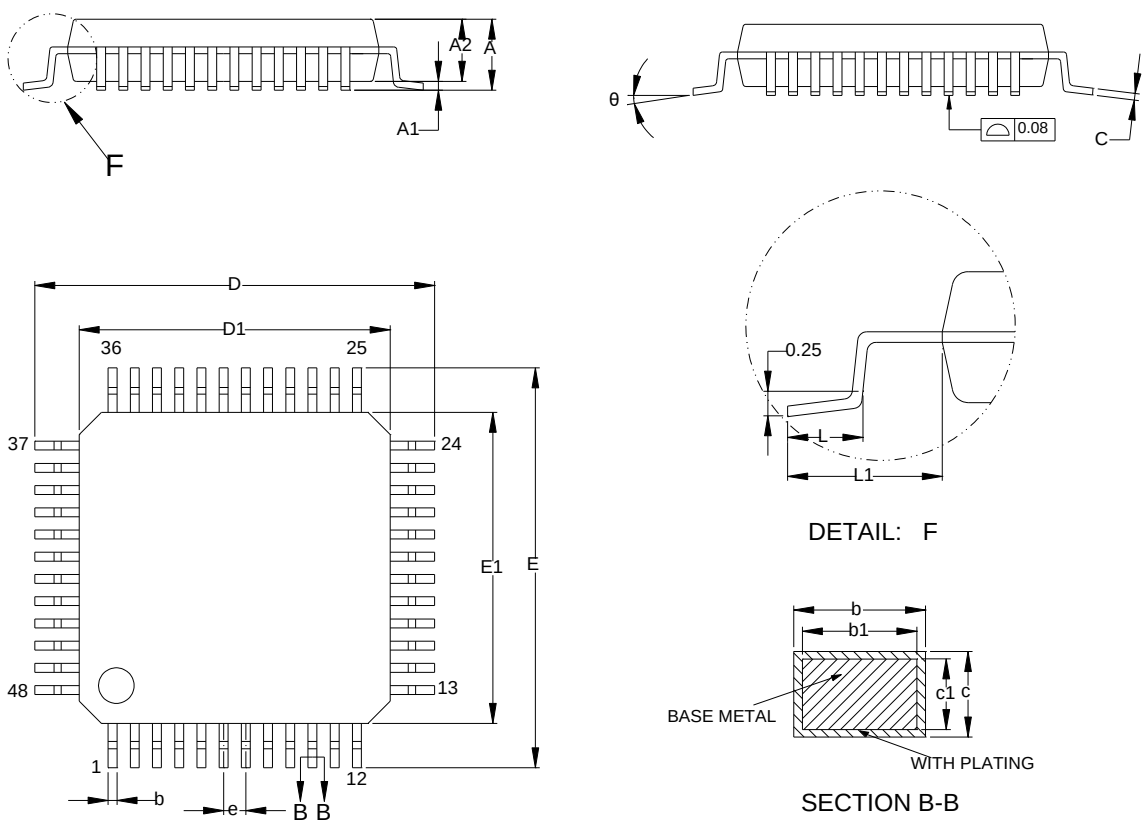
Pin	Name	Recommendation
1~16, 48	VC0~VC16	Cell inputs 1, 2, and 16 should always be connected to actual cells, with cells connected between VC1 and VC0, VC2 and VC1, and VC16 and VC15. VC0 should be connected through a resistor and capacitor on the pcb to pin 17 (VSS). Unused VC pins can be directly shorted to the adjacent VC pin. All VC pins should be connected to one of the following: an adjacent VC pin, an actual cell (through a resistor and capacitor), or the stack interconnect resistance (through a resistor and capacitor).
18, 20	SRP, SRN	If not used, these pins should be connected to pin 17 (VSS).
19, 44	NC	Pin 19 is reserved for OTP programming and should be connected to a 2.2 μF capacitor; leave it floating when OTP is not used. Pin 44 should be left floating.
21, 23, 25, 28, 29, 30, 31, 32	TS1, TS3, ALERT, SPI_MOSI, CFETOFF, DFETOFF, DCHG, DDSG	If not used, these pins can be left floating or connected to pin 17 (VSS).
22	TS2	If the device is intended to enter ShutDown mode, the TS2 pin should be left floating. If ShutDown mode will not be used in the application, and the TS2 pin will not be used for a thermistor or ADCIN measurement, the TS2 pin can be left floating or connected to pin 17 (VSS).
33	RST_SHUT	If not used, this pin should be connected to pin 17 (VSS).
34，35	REG1, REG2	If not used, these pins can either be left floating or connected to pin 17 (VSS) through a 0.1 μF capacitor.
36	REGIN	If not used, this pin should be connected to pin 17 (VSS) through a 0.1 μF capacitor.
37	BREG	If this pin is not used, it should be connected to pin 36 (REGIN).
38	FUSE	If not used, this pin can be left floating or connected to pin 17 (VSS).
39	PDSG	If not used, this pin should be left floating.
40	PCHG	If not used, this pin should be left floating.
41	LD	If the DSG driver will not be used, this pin can be connected through a series resistor to the PACK+ connector, or can be connected to pin 17 (VSS).
43	DSG	If not used, this pin should be left floating.
45	CHG	If not used, this pin should be left floating.
46	CP1	If not used, this pin should be connected to pin 47 (BAT). Note: In this case, enabling the charge pump will result in an additional current consumption of 800 μA.

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17 Package Information

17.1 Outline Dimensions

LQFP-48 Package Outline



NOTES:

1. All dimensions are in millimeters.
2. Refer to the Table 11 LQFP-48 dimensions (mm).

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Table 11. LQFP-48 dimensions (mm)

SYMBOL	MIN	TYP	MAX
A			1.60
A1	0.05	0.10	0.15
A2	1.35	1.40	1.45
D	8.90	9.00	9.10
D1	6.95	7.00	7.05
E	8.90	9.00	9.10
E1	6.95	7.00	7.05
C	0.10	0.15	0.20
c	0.09	0.15	0.20
c1	0.124	0.127	0.130
L	0.45	0.60	0.75
L1	1.00REF		
b	0.17	0.22	0.27
b1	0.17	0.20	0.23
e	0.50BSC		
θ	0°	3.5°	7°

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18 Device Comparison Table

PART NUMBER	Communications Interface	CRC	REG1 LDO Default
GD30BM2016CTTR/TY-I	I2C	N	Disable
GD30BM2016CTTR/TY -I01	SPI	Y	Disable
GD30BM2016CTTR/TY -I02	I2C	Y	Enable set to 3.3V
GD30BM2016CTTR/TY -I03	SPI	Y	Enable set to 5V
GD30BM2016CTTR/TY -I04	SPI	Y	Enable set to 3.3V

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19 Ordering Information

Ordering Code	Package Type	ECO Plan	Packing Type	MOQ	OP Temp.
GD30BM2016CTTR-I	LQFP-48	Green	Tape & Reel	3000	-40℃ to +125℃
GD30BM2016CTTR-I01	LQFP-48	Green	Tape & Reel	3000	-40℃ to +125℃
GD30BM2016CTTR-I02	LQFP-48	Green	Tape & Reel	3000	-40℃ to +125℃
GD30BM2016CTTR-I03	LQFP-48	Green	Tape & Reel	3000	-40℃ to +125℃
GD30BM2016CTTR-I04	LQFP-48	Green	Tape & Reel	3000	-40℃ to +125℃
GD30BM2016CTTY-I	LQFP-48	Green	Tray	250	-40℃ to +125℃
GD30BM2016CTTY-I01	LQFP-48	Green	Tray	250	-40℃ to +125℃
GD30BM2016CTTY-I02	LQFP-48	Green	Tray	250	-40℃ to +125℃
GD30BM2016CTTY-I03	LQFP-48	Green	Tray	250	-40℃ to +125℃
GD30BM2016CTTY-I04	LQFP-48	Green	Tray	250	-40℃ to +125℃

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20 Version History

Version Number	Description	Revision Date
1.0	Initial Release	5/5/2024

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